

One Session Treatment for Specific Phobias: An Adaptation for Paediatric Blood–Injection–Injury Phobia in Youth

Ella L. Oar¹ · Lara J. Farrell¹ · Thomas H. Ollendick²

Published online: 15 September 2015
© Springer Science+Business Media New York 2015

Abstract Blood–injection–injury (BII) phobia is a chronic and debilitating disorder, which has largely been neglected in the child literature. The present paper briefly reviews the aetiology of specific phobias with particular attention to BII and provides an integrated developmental model of this disorder in youth. Evidence-based treatments for child-specific phobias are discussed, and the development of a modified one session treatment (OST) approach to enhance treatment outcomes for BII phobia in children and adolescents is described. This approach is illustrated in two children with a primary diagnosis of BII phobia. The cases illustrate the unique challenges associated with treating BII in youth and the need for a modified intervention. Modifications included addressing the role of pain (e.g., psychoeducation, more graduated exposure steps) and disgust (e.g., disgust eliciting exposure tasks) in the expression of the phobia and fainting in the maintenance of this phobia. Moreover, it is recommended that parents be more actively involved throughout treatment (e.g., education session prior to OST, contingency management training, guidance regarding planning exposure tasks following treatment) and for families to participate in a structured e-therapy maintenance programme post-treatment.

Keywords Blood phobia · Injection phobia · BII · Children · Intensive treatment

Introduction

Specific phobia are one of the most common psychological disorders in children and adolescents, and they tend to have an early onset and are highly heterogeneous including animal (e.g., dogs, spider), natural environment (e.g., darkness, heights, water), blood–injection–injury (BII) (e.g., seeing blood, injections, being injured), situations (e.g., flying, elevators, enclosed spaces), and other (e.g., costume characters, illness, loud noises) subtypes. BII phobia is a particularly chronic and debilitating specific phobia that affects children, adolescents, and adults; however, it is one of the least well-understood subtypes. The disorder is characterised by fear and avoidance of seeing blood or an injury, or receiving an injection, or other invasive medical procedure (American Psychiatric Association (APA) 2013). Phobic avoidance of situations related to blood, injections, and injury is often associated with serious health consequences and impairment due to avoidance of seeing a doctor when ill, having immunisations, receiving medical treatment for diagnosed illnesses, and having operations and dental treatment (Marks 1988; Öst 1992; Öst and Hellström 1997). Moreover, individuals with BII phobia may also avoid certain career paths (e.g., nursing, medicine) and travel for fear of receiving necessary vaccinations (Öst and Hellström 1997; Öst et al. 1992). The disorder is associated with a complex clinical presentation and is thought to be distinct from other phobia types in that individuals often present with a unique physiological response (e.g., fainting). Moreover, the role

✉ Ella L. Oar
e.oar@griffith.edu.au

¹ School of Applied Psychology, Behavioural Basis of Health and Menzies Health Institute, Griffith University, Gold Coast Campus, Southport, QLD 4222, Australia

² Child Study Centre, Department of Psychology, Virginia Polytechnic Institute and State University, Blacksburg, VA 24060, USA

of disgust is implicated in BII phobia, a correlate shared only with small animal phobias, such as spider phobia.

To date, the clinical phenomenology, assessment, and treatment of BII phobia have largely been neglected in the child literature despite the fact that the disorder is not uncommon—affecting between 0.8 and 1.5 % of children and adolescents, and as many as 3–4.5 % of adults. Retrospective reports of adolescents and adults suggest that BII phobia typically has its onset prior to 10 years of age (Bienvenu and Eaton 1998; Burstein et al. 2012; Essau et al. 2000; Öst and Hellström 1997), highlighting the persistent nature of the disorder and the need for interventions during childhood. Surprisingly, however, there are no evidence-based treatments for youth with BII phobia at this time. Therefore, the present paper aims to briefly review the aetiology of specific phobia with a focus on BII phobia and to propose an integrated developmental model of BII phobia in children and adolescents. The current status of treatments for youth is discussed, and case illustrations are used to describe the development of a case formulation-driven, intensive cognitive behavioural treatment for BII phobic youth.

Vulnerability Factors Associated with Specific Phobia

Specific phobias have a complex aetiology that is multi-determined (King et al. 2004). Genetic influences, aberrant brain processes, temperament, parenting, learning experiences, evolutionary preparedness, cognitive biases, and avoidance are all factors thought to be involved in the development and maintenance of a specific phobia (Cowart and Ollendick 2013; Ollendick and Muris 2015). In addition to the above, BII phobia is proposed to have a number of unique factors, in comparison with other phobia subtypes, that are implicated in its onset and course. Vulnerability factors associated with specific phobia in childhood are reviewed, highlighting the degree to which these factors may be implicated in BII phobia.

Evolutionary Preparedness

The non-associative model of fear acquisition proposes that some fears, such as a fear of heights, snakes, spiders, and blood, are biologically prepared for through evolution (Menzies and Clarke 1995). According to this model, these fears do not require critical learning experiences because they have been passed on to us from our ancestors (e.g., genetically encoded) in as much as they were evolutionarily adaptive and necessary for survival. For example, a fear of blood may be characterised as an evolutionarily adaptive fear. The model, however, has been criticised (Muris et al. 2002), as it cannot on its own account for the

development of a specific phobia. Rather it has been suggested that we inherit a vulnerability to be vigilant towards these objects and stimuli (e.g., blood, spiders, heights); however, in some children and adults, this vigilance and fear becomes pathological in nature through a complex interaction of biological vulnerability, temperament, and social learning experiences (Muris et al. 2002).

Genetics

Family and twin studies have found a genetic contribution to the development of specific phobia. First-degree relatives of adults with a specific phobia are at significantly higher risk of developing their own specific phobia (31 %) in comparison with the first-degree relatives of controls with no psychiatric diagnoses (11 %; Fyer et al. 1990). Moreover, children of parents with a specific phobia have been found to be at increased risk for the development of a specific phobia but only for the phobia displayed by their parent (Fredrikson et al. 1997; Fyer et al. 1995). In relation to BII phobia specifically, Öst (1992) reported that 61 % of blood injury phobic adults and 29 % of injection phobic adults reported having a first-degree relative with the same fear, which was significantly higher than other phobia types (e.g., animal, dental, and claustrophobics). Moreover, Van Houtem et al. (2013) recently conducted a review and meta-analysis of 10 adult twin studies to examine differences in the heritability of the varying phobia subtypes. The highest heritability rates were found for BII phobia ranging from 28 to 63 %, followed by animal phobia at 22 to 44 %. Thus, BII phobia is often present in parents as well as their children, perhaps more so than in other phobia subtypes.

Neurobiology

Neuroimaging studies over the past 20 years have examined area-specific changes in metabolic activity in the brain which underlie aberrations in the processing of fear. In a recent review, Del Casale et al. (2012) concluded that phobic individuals show abnormal activation of brain areas, such as the amygdala, anterior cingulate cortex, thalamus, and insula—all of which are associated with the emotional perception and early amplification of fear. Moreover, they found that following exposure to phobic stimuli, brain areas in the prefrontal cortex, which regulate fear, are less activated in individuals with a specific phobia in comparison with healthy controls. Hence, in terms of brain structure, sufferers of specific phobia appear to be more easily and strongly aroused when confronted with phobic stimuli and are less able to regulate their arousal. Across the phobia types there appear to be shared neural correlates and also some distinct neural substrates relevant

to the different phobia types. Caseras et al. (2010) compared the neural responses of BII phobic adults ($n = 12$), spider phobic adults ($n = 14$), and healthy controls ($n = 14$) during the presentation of phobia-relevant stimuli using functional magnetic resonance imaging (fMRI). BII adults showed increased activation in the thalamus and visual/attention areas (occipito-temporo-parietal cortex) in comparison with the spider phobic adults and healthy controls. While providing interesting findings in relation to the brain circuits involved in specific phobia, neuroimaging studies have predominantly been carried out with animal phobic adults (particularly spider subtype); therefore, further research is needed in order to investigate fear processing across the different phobia subtypes and, particularly so, in youth.

Temperament

Existing research suggests that temperament is associated with the development of a specific phobia (Ollendick and Muris 2015). Behavioural inhibition is a temperamental characteristic, which is described as a tendency to demonstrate distress, fear, and avoidance when confronted with unfamiliar situations, persons, and objects. It is believed to be inherited, is observed in 10–15 % of children (Kagan et al. 1984), and appears to be a risk factor for the development of both phobic and anxiety disorders (Kagan et al. 1984). Biederman et al. (1990, 1993) found that behaviourally inhibited young children (2–8 years) were more likely to have anxiety disorder, including specific phobia, in comparison with uninhibited children. Moreover, Muris et al. (1999) examined the relationship between self-reported behavioural inhibition in a community sample of older children ($N = 152$; 12–14 years) and anxiety and depressive symptoms. Youth were provided with a definition of behavioural inhibition and were asked to classify themselves as low, medium, or high in this temperamental characteristic. Following this, they completed a range of self-report measures including scales assessing symptoms of specific phobia (e.g., animal, environmental, BII and situational). In comparison with the low- and middle-behavioural-inhibition groups, children in the high-behavioural inhibition group endorsed specific phobia symptoms more frequently. The degree to which behavioural inhibition is associated with BII phobia in children and adolescents, relative to the other phobia subtypes, has not yet been explored.

Parenting

Parenting practices and parental psychopathology are also believed to be involved in the development and maintenance of childhood anxiety disorders including samples of

youth with specific phobias. Parents of anxious youth have been found to have a more intrusive and overprotective parenting style (Breinholst et al. 2012; Gar and Hudson 2009). Parents with this style may prevent their child from engaging in age-appropriate activities and responsibilities. They are also more likely to intervene and attempt to protect their child from new, difficult, or anxiety-provoking situations. In regards to phobic youth, this may involve accommodating and reinforcing their child's phobic avoidance. Hence, children do not have the opportunity to develop coping skills in fear-inducing situations, and also do not gather evidence that disconfirms their threat expectancies and low self-efficacy in dealing with their fears (Bögels and Brechman-Toussaint 2006; Breinholst et al. 2012; Hudson and Rapee 2001, 2002). Parental anxiety and other psychopathology can also influence the development and maintenance of childhood-specific phobia through modelling and reinforcement of anxious and avoidant behaviour, which can be accounted for by social learning experiences. The actions and instructions given by an anxious parent in anxiety-provoking situations may also convey to the child that the situation is threatening and therefore needs to be avoided (Bögels and Brechman-Toussaint 2006). Existing research has found that anxious children, in comparison with non-anxious children, tend to have parents who encourage avoidance (Dadds et al. 1996; Ginsburg et al. 2004). Unfortunately, these studies have been conducted with transdiagnostic anxious samples with no studies to date specifically examining parental rearing factors in youth with specific phobia or BII.

Learning

According to Rachman (1976, 1977), there are three learning pathways associated with the development of a specific phobia: direct/classical conditioning, vicarious conditioning (modelling), and negative information transmission. Classically conditioned phobias are acquired through a direct negative experience with the phobic object/situation. Fear develops when a neutral stimulus becomes associated with an aversive outcome (unconditioned stimulus), such that the neutral stimulus takes on its aversive properties (becoming a conditioned stimulus). The likelihood of a conditioned fear developing is dependent upon the intensity of fear/pain experienced in the presence of the stimuli and/or the frequency and repetition of the association between the conditioned stimulus and fear/pain. An example of a direct/classically conditioned phobia would be a child who has a painful vaccination and goes on to develop an injection phobia. Consistent with this, many BII phobic adults report that a painful experience preceded the onset of their fear (Öst 1991, 1992). The direct/classical conditioning pathway to phobia acquisition assumes that a

person has had a direct negative experience with the phobic stimulus; however, this is not always the case and individuals can develop a phobia despite having no direct contact with the phobic object. In these individuals, indirect pathways play a crucial role in fear acquisition. A phobia may be acquired vicariously through modelling and observation of other's anxious behaviour. An example of this type of phobia acquisition would include a child who develops a blood phobia after watching their mother display a fearful reaction while having a blood test. Negative information transmission is the third learning pathway proposed to be associated with specific phobia acquisition. A child may acquire a phobia through hearing or reading negative information about an object/situation. For example, a child might develop an injection phobia after hearing a peer talking about being held down and forcibly given an injection. Evidence for these pathways has been established in children with specific phobia (King et al. 1997; Merckelbach et al. 1996), as well as adults with BII phobia (Öst 1991); however, this pathway has not been directly assessed among youth with BII.

Cognitive Biases

A large body of research has found that various cognitive biases are associated with the development and maintenance of specific phobias. Experimental studies with children and adults have demonstrated that phobic and anxious individuals are hypervigilant to threatening stimuli and allocate their attention towards threatening stimuli relative to neutral stimuli (Bar-Haim et al. 2007; Cisler and Koster 2010). Furthermore, these individuals may have more difficulty disengaging their attention away from threatening stimuli (Fox et al. 2001). This phenomenon is known as attentional bias. These biases are activated when initially confronted with a feared object or situation, and as such, have a distorting effect on further information processing (Muris and Field 2008). Attentional biases have been observed in both anxious adults (Armstrong et al. 2013; Mogg and Bradley 2006) and anxious children (Waters et al. 2014a, b). In regards to adults with BII phobias, findings relating to attentional biases have been mixed (Armstrong et al. 2013; Haberkamp and Schmidt 2014). There is some evidence to suggest that BII phobia may be associated with a vigilance avoidance pattern whereby individuals are initially vigilant and then rapidly disengage from the stimuli. Hence, individuals with BII phobias may exhibit shorter fixations or glances towards threat (Mogg et al. 2004). In a recent study, Armstrong et al. (2013) found that high injection fearful ($n = 33$) adults oriented to injection-related images more often than low injection fearful ($n = 32$) adults. However, they did not orient to injection images more frequently than other emotional

images (e.g., attack, appetitive and neutral). Consistent with a vigilance avoidance pattern, high injection fearful adults rapidly disengaged from injection images after initially seeing them. Overall, they also looked at the injection images significantly less frequently than the other emotive images. This pattern was not observed in the low injection fearful group. Finally, attentional avoidance of injection threat was found to predict avoidance during an injection behavioural avoidance task (BAT). A weakness of existing literature in the area is that studies have predominantly used BII fearful samples, determined by the use of self-report data only. Few studies have examined attentional bias in BII phobic adults who have been diagnosed using structured clinical interviews.

Only a few studies exist that have examined attentional bias in children with specific phobia. Waters et al. (2014a) investigated differences in attentional biases in 435 children (5–13 years) who had either a principal fear disorder, including specific phobia, social phobia and separation anxiety disorder ($n = 158$) or a principal distress disorder which included generalised anxiety disorder ($n = 75$), as well as a healthy control group ($n = 202$). Youth with a principal fear diagnosis (e.g., specific phobia) were found to have a significant bias away from threat faces relative to neutral faces when compared to those in the principal distress disorder and healthy control groups. Attentional biases in youth with BII specifically have not yet been examined.

Other cognitive biases, such as interpretation biases, have also been implicated in the development and maintenance of specific phobias. Children and adults with specific phobia have been found to overestimate or inflate the association between phobia-related stimuli and negative outcomes. Jones and Menzies (2000) examined perceptions of danger in 15 spider phobic adults and 15 controls before, during, and after a spider avoidance task. Spider phobic adults rated higher estimates of being bitten by the spider and more severe consequences resulting from a spider bite in comparison with controls. Harm beliefs were found to predict avoidance during the spider avoidance task. Similar cognitive distortions have been reported across other phobia subtypes (de Jongh et al. 1995; Marshall et al. 1992; Menzies et al. 1998; Rachman and Cuk 1992).

Limited knowledge exists regarding the role of cognition in children with specific phobia. Ollendick et al. (2010) reported no differences in danger expectancies between animal and natural environment phobic youth, with both phobia types reporting similarly high levels of the probability of harm associated with feared stimuli and low expectancies for coping with negative phobia-related outcomes. Byrne et al. (2014) investigated whether harm beliefs in dog phobic children ($N = 27$) predicted

avoidance and distress before and after exposure therapy. Children were shown a live dog and asked to rate how much they believed the dog would harm them (e.g., the dog would bite them or jump on them). Harm beliefs predicted distress during a pre-treatment BAT and avoidance during the post-treatment BAT. In a recent study, Oar et al. (submitted) found that youth with BII phobia ($n = 27$) reported more exaggerated danger expectancies and more often had a focus of fear on physical symptoms (e.g., faintness and nausea) in comparison with youth with animal phobia ($n = 25$).

Perceived Control

Perceived control, which is defined as the belief one has about the nature of control over situations and events, has been found to play a role in the manifestation of childhood anxiety and phobic disorders (Weems and Silverman 2006). It consists of two dimensions (1) the location of the control (e.g., internal, external) and (2) the situation or object the control is over (e.g., health, emotional reactions, bodily sensations, threatening stimuli and interpersonal relationships). Weems et al. (2003) examined the role of control beliefs in anxious youth ($n = 86$) with a primary diagnosis of generalised anxiety, social phobia, separation anxiety disorder, or specific phobia in comparison with non-clinical controls ($n = 31$). Perceived control over feared events was significantly and negatively correlated with self-reported anxiety symptoms in both the anxious and non-clinical samples. Of note, however, children with anxiety disorder diagnoses endorsed significantly lower perceived control than the non-clinical controls. In a recent study, Kane et al. (2014) investigated the relationship between fearfulness, anxiety sensitivity, maternal over protection, and perceived control in a sample of 126 (7–13 year old) youth with a primary diagnosis of specific phobia, excluding BII phobia. Control beliefs were found to be a significant predictor of children's fearfulness, with heightened levels of fear associated with lower perceived control. Subsequent analysis revealed that control beliefs were a significant predictor of lower perceived control in younger children, however not older youth, and for girls but not boys. Kane et al. (2014) suggested that younger children may have less-developed cognitive capacities, which limit their ability to understand anxiety-provoking situations and their control over such situations. Additionally, the authors commented on the socialisation of boys in society who are taught that they should be able to manage and suppress fear, hence encouraging them to express control over anxiety-inducing events. Finally, this study revealed that lower perceived control in anxiety-related situations was significantly predictive of greater levels of fear in children with a specific phobia and

comorbid anxiety disorders; however, it was not a predictor of youth with a phobia diagnosis alone. It was suggested that youth with multiple anxiety disorder diagnoses would experience lower feelings of control in their day-to-day life in comparison with the phobia-alone youth, who may experience this lack of control only in relation to their specific feared object or stimuli. Given that for many youth with injection fears, the feared outcome (i.e., pain) is under external control (i.e., medical professional), and influenced by health conditions also outside the child's control; it is likely that perceived control may be particularly relevant to BII phobia in children and youth. While the role of perceived control has not been empirically examined in youth with BII, it is noteworthy that clinically, children with injection fears (in our experience) report a preference for watching the injection, or for giving themselves a finger prick rather than the therapist doing this to the child, which highlights the child's need for increased control during the experience.

Vulnerability Factors Associated with BII Phobia

It is evident from the literature reviewed that there are a host of vulnerability factors (e.g., evolutionary preparedness, genetics, neurobiology, temperament, parenting, learning experiences, and cognitive bias) that increase a child or adolescent's risk of developing an anxiety or phobic disorder. A growing body of research with adults suggests that in addition to the above, BII phobia may have a number of unique factors (e.g., fainting, disgust, physiological symptoms, and pain), in comparison with other phobia subtypes, that are implicated in its onset and maintenance.

Fainting

An important feature distinguishing BII phobia from other phobia types is its relatively unique physiological response (Öst 1991), with many BII phobic adults (56–100 %) reporting a history of fainting when confronted with their phobic stimuli (Ayala et al. 2009). Comparatively, only 0.02 % of adults with other phobia subtypes (Connolly et al. 1976; Öst 1992; Thyer et al. 1985) and 2 % with other anxiety disorders (Connolly et al. 1976) report fainting in the presence of their feared stimuli. In the literature, this fainting response has been described as diphasic, with an initial phase involving a rapid acceleration of heart rate and blood pressure, as is typical of the fight or flight anxiety response, followed by a second phase which is characterised by a sharp decrease in heart rate and blood pressure (Thyer et al. 1985). This rapid decline in heart rate and blood pressure leads to reduced cerebral blood flow, and ultimately can result in fainting (Ayala et al. 2009). More

recently, researchers have begun to look beyond the cardiovascular changes associated with BII fainting (Ritz et al. 2010) and have examined additional physiological parameters, such as changes to respiration including hyperventilation, that may also have a role in this response pattern (Lumley and Melamed 1992). Recently, Oar et al. (submitted) compared the phobic response of youth with BII phobia ($n = 27$) to those with dog phobia ($n = 25$). Five youth in the BII phobia group reported a history of fainting in the presence of their feared stimuli, with a further five reporting a history of vomiting in the presence of feared stimuli. Comparatively, no youth in the dog phobia group reported a history of either fainting or vomiting.

The precise relationship between BII phobia and a fainting response is not yet fully understood. Consistent with the non-associative fear model, some researchers have suggested that fainting is an adaptive response to threat. Slowing of heart rate and blood pressure in response to blood or injury produces immobility and this may be an adaptive reflex with predators losing interest in a prey if it is immobile (i.e., playing dead; Marks 1988). Alternatively, Barlow (2004) suggested that our ancestors responded with a significant drop in blood pressure to minimise blood loss and the danger of shock, helping us to be more likely to survive an attack or injury. However, evolutionary theories of BII fainting have received criticism and cannot fully account for all aspects of the response (Page 1994). Early research conducted by Page and Martin (1998) suggested that individuals with BII phobias may inherit a tendency to faint in non-blood-related situations as well. In a multivariate genetic analysis involving 659 twin pairs, they found that BII fainting was explained by additive genetic variance predictive of fainting in non-blood situations, and also unique environmental experiences related to fainting in the presence of blood. Page and Tan (2009) proposed that the development of BII phobia may involve a process whereby aversive experiences associated with fainting directly condition fear to BII situations. Moreover, they hypothesise that disgust plays a key role in BII-related fainting, and that when disgust and fear are elicited by the same stimuli (e.g., BII stimuli), this may increase the likelihood of fainting. However, the fainting response cannot be completely explained by the co-occurrence of the two emotions. Page and Tan (2009) hypothesise that the type of stimuli, disgust response involved (state versus trait disgust), a predisposition to faint, and appraisal of the stimuli are all involved in the complex interaction that results in fainting.

Interestingly perceived control may also be associated with fainting in BII phobia. Little is known about the role of control cognitions in the vasovagal response. However, in a recent study, Gilchrist et al. (2015) examined the effects of experimentally manipulated perceived control

during a task (e.g., watching a surgery video) demonstrated to induce a vasovagal responding in non-medical personnel. Eighty-two university students (18–30 years) watched a 10-min video of an open heart mitral valve surgery and were randomly assigned to either (1) being able to take a 2-min break whenever they requested during the video (e.g., perceived control group) or (2) were informed that the experimenter would assign them a 2-min break during the task. Those in the no-perceived control group reported significantly greater vasovagal symptoms and anxiety. The findings were corroborated with physiological data showing lower stroke volume, cardiac output, and diastolic blood pressure. Moreover, for participants who indicated they were fearful of blood, enhancing their sense of perceived control reduced their vasovagal symptoms. It appears that perceived control may indeed be implicated in the fainting response and may be an important target for enhancing treatment outcomes in BII phobia. Further research is needed to better understand the role of fainting in BII phobia and the interplay between fear and disgust, as well as perceived control especially in children. To date, studies examining the fainting response associated with BII phobia have almost been exclusively carried out with adult samples; thus, it is currently unclear whether fainting is a unique correlate of BII phobia children and adolescents. Moreover, there is some evidence that vomiting may be associated with the disorder in youth (Oar et al., submitted).

Disgust

Traditionally, BII phobia has been associated with fear and anxiety; however, in recent years, there has been considerable research into the important role of disgust in the aetiology and maintenance of the disorder (Ayala et al. 2009; Öst 1992; Page and Tan 2009). Numerous studies have found that self-report measures of disgust and disgust sensitivity are positively associated with measures of BII phobia (Phillips et al. 1998). Disgust sensitivity is the trait like predisposition to experience disgust towards a range of aversive stimuli (Moretz et al. 2011; Olatunji and Cisler 2009). Disgust sensitivity has been found to be positively correlated with neuroticism and negatively with sensation seeking. de Jong and Merckelbach (1998) demonstrated that BII and spider fearful adults show domain specificity among disgust elicitors. They found significant correlations between measures of BII phobia and disgust associated with bodily wounds and punctures and between measures of spider phobia and animal-related disgust elicitors. Moreover, existing research suggests that BII fearful adults not only do report high levels of disgust and aversion towards blood and mutilation stimuli, also experience heightened disgust responses towards a range of disgust elicitors that are unrelated to their fearful stimuli (Olatunji

et al. 2007). For example, Sawchuk et al. (2000) and Tolin et al. (1997) found that BII fearful adults reported significantly higher levels of disgust towards odours, rotting foods, and bodily products (e.g., mucus, vomit, faeces) than non-fearful individuals. Additionally, BII fearful adults reported greater levels of disgust while watching videos of maggots (Sawchuk et al. 1999) and sewage (Sawchuk et al. 2002) than non-fearful adults. Together these studies suggest that BII phobia is characterised by a heightened generalised disgust sensitivity (Olatunji et al. 2007a) which could be involved in the development and maintenance of this disorder.

In children, disgust sensitivity has been found to be significantly correlated with small animal and spider phobias, as well as BII phobias (de Jong and Muris 2002; Muris et al. 2008). More specifically, Muris et al. (1999, 2008) found that disgust sensitivity was associated with BII phobia symptoms in samples of non-clinical children, even after controlling for the effects of trait anxiety and neuroticism. As previously discussed, the majority of research investigating the relationship between disgust and specific phobia has focused on adults, while relatively little attention has been given to child samples, and to date there are no clinical studies with children.

Although anxiety certainly appears to play an important role in BII, some research suggests disgust may play a more pivotal role than anxiety. Research conducted by Tolin et al. (1997) and Sawchuk et al. (2002), for example, has found that when confronted with their phobic stimuli, BII fearful adults report greater disgust than fear. Tolin et al. (1997) presented BII fearful ($n = 24$), spider fearful ($n = 59$), and non-fearful ($n = 74$) controls with colour photographs of spiders and injections. During the tasks, participants completed fear and disgust emotional rating forms. As expected, BII fearful adults reported significantly higher levels of fear and disgust when confronted with the injection photograph than spider and non-fearful participants. Moreover, BII fearful adults reported higher levels of disgust than fear when viewing phobia-related stimuli, whereas spider fearful adults reported equal levels of fear and disgust. Sawchuk et al. (2002) replicated these findings whereby BII fearful adults ($n = 37$) reported significantly higher levels of fear and disgust relative to controls, and furthermore, BII fearful adults reported higher levels of disgust than fear.

Physiological Symptoms/Pain

The physiological symptoms experienced when confronting their phobic stimuli is often a significant source of distress for adults with BII phobias and can perpetuate their avoidance of BII-related stimuli (Öst et al. 1984b; Page 1994; Ritz et al. 2010). In comparison with other phobia

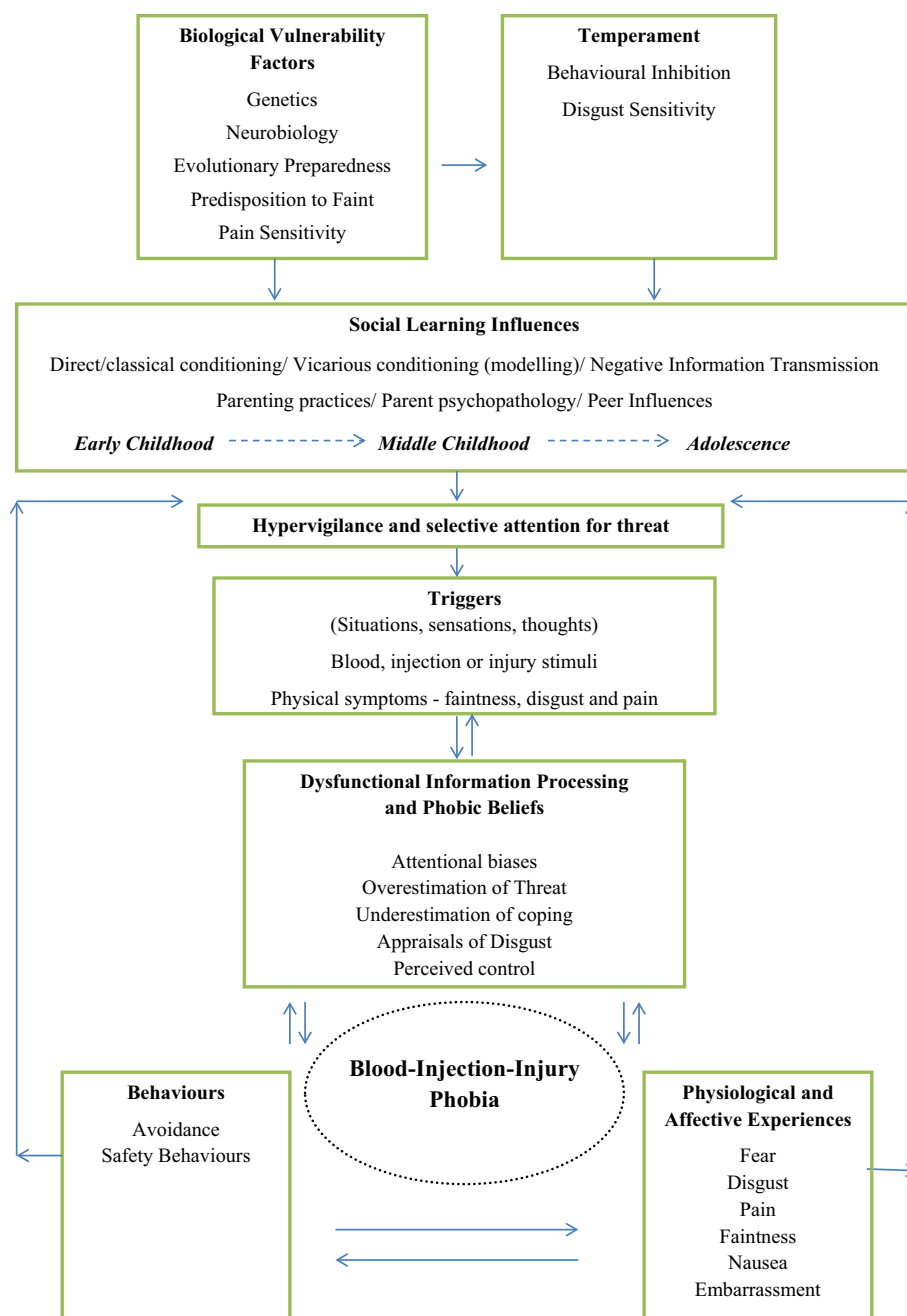
subtypes, studies have found that BII phobic adults tend to focus their cognitions on physical symptoms (e.g., fainting) and internal feelings (e.g., disgust and revulsion) and to a lesser extent on danger and harm cognitions (Antony et al. 1997; Lipsitz et al. 2002). BII phobics may also fear the secondary consequences of their physiological response such as embarrassment associated with fainting or worry about being injured when fainting (Öst 1992; Page 1994). Moreover, research suggests that BII fears may be maintained by a tendency to exaggerate pain which is then consequently associated with increased pain intensity (Severeijns et al. 2001). This exaggerated pain response and elevated pain intensity are believed to perpetuate phobic avoidance. In the adult literature, BII and dental fearful individuals have been found to have a tendency to overestimate pain and to endorse a lower perceived ability to tolerate physical pain and discomfort (Arntz et al. 1990; Smith and Meuret 2012). The role of pain and physiological symptoms in the development and maintenance of BII fears in children is yet to be examined.

An Integrated Developmental Model of BII Phobia

Based on our review of the literature, an integrated developmental model of BII phobia in children and adolescents is proposed (see Fig. 1). This model incorporates and draws upon the existing aetiological pathways for childhood-specific phobias more generally (Muris and Merckelbach 2001; Ollendick and Muris 2015) and adult BII phobia more specifically (Page and Tan 2009). In this model, it is proposed that BII phobia in children and adolescents has a complex aetiology that is multi-determined. Evolutionary preparedness, genetics, neurobiology, temperament, parenting, learning experiences, cognitive biases, perceived control, fainting, disgust, physiological symptoms, and pain are all hypothesised to be involved in the development and maintenance of the disorder in youth.

Several assumptions, derived from the adult BII phobia literature, have been included in the model due to the limited research conducted in paediatric samples. These assumptions, however, need to be empirically examined in children and adolescents and differences in the developmental expression of BII considered. Transient fears are a normal part of child development (Gullone 2000; Muris et al. 2000) and the content of such fears tend to follow a predictable course that is associated with increasing cognitive ability. In infancy and toddlerhood children typically experience concrete fears of stimuli in their immediate environment (e.g., fear of strangers, animals etc.); however, in middle childhood, fears become increasingly more abstract (e.g., ghosts, the supernatural) and this continues into adolescence (e.g., social fears). Specific fears are less frequent over the course of childhood, peaking in early

Fig. 1 A proposed integrated developmental model of BII-specific phobia



childhood between the ages of 7–9 years and declining in children 10 years and older (Muris et al. 2000). However, some research suggests that medical fears may be an exception to this and in fact may increase with age (Gullone 2000). Indeed, Taddio et al. (2012) report that injection-related fears are highly prevalent in both children and teenagers, affecting between 50 and 68 % of all youth. The developmental progression of fears therefore needs to be taken into consideration when assessing BII phobia in children and adolescents, with bodily injury and medical fears normative to some degree. However, for a subset of

youth these normal fears may become more frequent, intense, durable, and impairing eventually evolving into a phobia (Milliner et al. 2013). While many children may become highly distressed immediately before having an injection, or at the time of an injury, key indicators that might differentiate normal fears from pathological fear might include (a) the degree to which children become preoccupied by the fear (e.g., asking regularly about having a vaccination or medical check-up), (b) excessive avoidance of feared stimuli (e.g., complete refusal to see a medical professional), and/or (c) the degree of interference

to the child or families life (e.g., children being unable to enrol in school due to absence of vaccinations; family not holidaying for fear of vaccinations).

Within BII phobia, differences in the expression of symptoms may be present from early childhood through to late adolescence depending on developmental stage. For example, younger children may present with exaggerated behavioural expression of fear and pain (e.g., kicking, screaming, biting), with parents reporting children need to be restrained for injections (LeBaron and Zelter 1984). However, in comparison, anxiety symptoms may be less apparent in adolescents, who may present as more withdrawn, sad, irritable, or angry when faced with their feared stimuli.

Factors involved in the maintenance of BII phobia may also vary at different developmental stages. In early and middle childhood, parenting practices and parent psychopathology may play a stronger role in the development and maintenance of BII phobia. However, for adolescents, who often receive vaccinations in school-based programs, the role of parents in the maintenance of their fear may be less important. In the broader child anxiety literature, findings relating to the influence of parent factors, child age, and child anxiety have been mixed, with some models proposing that parent influences decrease as children enter adolescence, while others models postulate that the influence of parents on child anxiety increases overtime due to its cumulative effects (Wei and Kendall 2014). Clinically, it has been observed in our experience, and consistent with the normal progression of developmental fears (Gullone 2000), that adolescents with BII may report secondary anxiety associated with their fear of injections and blood, including fear of embarrassment related to their phobic response (e.g., fainting and losing control); hence, peer influences and social anxiety may be more influential at this age.

Finally, it is important to recognise that developmental differences may also occur with regards to treatment-seeking behaviour and motivation to engage in treatment. Young children with BII may have very limited motivation to overcome their fear in comparison with adolescents. Adolescents on the other hand have greater cognitive capacity to understand the health-related benefits of injections and medical intervention, in addition to having the ability to more rationally process threat associated with blood and injury, and therefore may be more willing to engage in therapy. Adolescents may also possess greater insight into the impact of their BII-related fears on future career options (e.g., nursing, veterinary science, childcare worker) and thus may be more motivated to overcome their fear. In our experience, adolescents often play a part in initiating treatment for their fears, whereas for younger

children, parents typically initiate and seek treatment for their often reluctant children.

Addressing components of the proposed model is hypothesised to lead to a better understanding of BII phobia in youth and assist in the development of individualised, developmentally sensitive, case formulation-driven approaches to assessment and treatment. The model highlights the importance of the assessment of the children's unique phobic response (e.g., fainting), including the role of pain, disgust, threat beliefs, parent psychopathology, and the influence of familial and peer processes in the maintenance of their child's disorder. A hypothesis testing approach can be used to identify the key maintaining mechanisms for each youth, and following this, treatment components can be matched to this formulation in order to attempt to maximise treatment outcome. Following a brief review of the treatment of specific phobias in youth, this model is described further and illustrated with two case studies.

Treatment for Specific Phobia in Youth

Cognitive behavioural therapy (CBT) is considered an evidenced-based treatment for paediatric specific phobia. To date, four large randomised controlled trials (RCT; Ollendick et al. 2009, 2015; Öst et al. 2001; Silverman et al. 1999) and seven smaller clinical trials (Farrell et al. 2013; Flatt and King 2010; Leutgeb et al. 2012; Leutgeb and Schienle 2012; Muris et al. 1997, 1998; Waters et al. 2014b) support its effectiveness. These trials have compared CBT to waitlist control (Flatt and King 2010; Leutgeb et al. 2012; Leutgeb and Schienle 2012; Öst et al. 2001), an education support group (Ollendick et al. 2009; Silverman et al. 1999), and other psychological treatments (e.g., eye movement desensitisation and reprocessing, psychoeducation, cognitive therapy; Flatt and King 2010; Muris et al. 1997, 1998). Traditionally, CBT for specific phobia involves 8–12 weekly sessions (approximately 1 h each in duration). However, in the 1980s seminal work by Öst led to the development of an intensive treatment approach, called one session treatment (OST), that involves a single session of massed exposure that is maximised to 3 h in duration (Milliner and Farrell 2014; Öst 1989). OST was initially trialled with adults, delivering favourable outcomes (Öst et al. 1991a, 1997), and following this was successfully adapted and tailored for use with children and adolescents. During the single intensive session, a range of cognitive behavioural techniques are utilised, including exposure, cognitive challenges, participant modelling, reinforcement, and psychoeducation (see Davis III et al. 2012 for a through treatment description). To date, most published treatment studies for paediatric specific phobia have used an OST approach which has resulted in it now

being considered a *well-established* treatment for phobic youth, based on the stringent criteria developed by Chambless et al. (1998; Chambless and Ollendick 2001).

Current Status of Treatment for BII Phobia in Children and Adolescents

Across the published studies, children with a diverse range of specific phobias (e.g., animal, natural environment) have been included. Notably, however, children with BII phobias have been underrepresented in many samples and excluded in others. Moreover, to date, only a few single-case studies have been published (Sanders and Jones 1990; Thompson 1999), in addition to the small numbers of youth with BII who have been included in the RCTs (Flatt and King 2010; Öst et al. 2001; Silverman et al. 1999). In one of the case studies, Sanders and Jones (1990) treated a 13-year-old girl who required major surgery and suffered from phobias of injections, dental, and medical procedures. The adolescent also presented with comorbid oppositional defiant disorder. Treatment consisted of 21 sessions involving coping skills training, systematic desensitisation, imaginal and in vivo desensitisation with participant modelling and daily home tasks. Following treatment the adolescent displayed significant improvements on measures of anxiety including ratings of distress, performance on a BAT, and responses on self-report measures. Moreover, these gains were maintained at an 8-month follow-up. In the second case study, Thompson (1999) evaluated the effectiveness of CBT for a 14-year-old girl with a BII phobia who had a history of fainting at the sight of blood. The adolescent completed 13 one-hour sessions, which included exposure, applied tension, and cognitive restructuring. Results showed substantial reductions in subjective distress in BII situations from baseline to post-treatment. On phobic and general self-report measures of anxiety, reductions were also observed and, furthermore, fainting did not occur during treatment. The adolescent reported that she found the cognitive restructuring component of treatment the most helpful.

In the RCTs, Öst et al. (2001) included 12 youths with injection phobias and two with blood phobias in their trial of 60 youth, which evaluated the efficacy of an intensive OST for specific phobia in children and adolescents. They found that BII phobic youth responded significantly less favourably to treatment than youth with other types of phobia during a post-assessment BAT. Based on their clinical impression from working with the children, Öst et al. (2001) hypothesised that the youth had difficulty differentiating the therapist from other health professionals (e.g., doctor, nurse) who they associated with previous anxiety-provoking medical experiences. Hence, they were

less likely to engage in therapist-suggested exposure tasks. Silverman et al. (1999) and Flatt and King (2010) also included a small number (4–6 participants) of youths with BII phobias in their controlled treatment trials; however, they did not examine differences in treatment outcome across the different types of phobia. In the two largest RCTs conducted to date, Ollendick et al. (2009) specifically excluded youths with BII phobias due to their unique physiological presentation (e.g., fainting), poorer treatment response (Öst et al. 2001), and the challenges associated with delivering treatment in an outpatient setting (i.e., need for medical professionals to assist in exposure). Moreover, in two recently published RCTs for youth with specific phobias, Waters et al. (2014b) and Farrell et al. (2013) youth with BII phobias were excluded.

Current Status of Treatment for Adult BII Phobia

Behavioural and cognitive-behavioural procedures have received considerable empirical support for the treatment of BII phobia in adults (Ayala et al. 2009). In-vivo exposure and applied tension have been found to be particularly effective. Applied tension is a coping technique that can be implemented during exposure to BII stimuli (Ayala et al. 2009; Öst and Sterner 1987). The technique involves brief tension of arms, legs and torso muscles until the person feels warmth rising in their face (10–20 s) followed by 20–30 s of release of the muscles. The procedure is repeated up to five times. Applied tension is proposed to raise blood pressure, venous return, and cerebral blood flow and hence reduce the likelihood of fainting. To date five controlled treatment trials have been conducted for adults with BII phobia (Hellström et al. 1996; Öst et al. 1984a, 1989, 1991b, 1992), all conducted by the one research group, led by Öst and colleagues from Sweden. Interventions used in the studies included massed or spaced exposure, applied tension, tension only, applied relaxation, and combination of applied tension and applied relaxation. In a recent review, Ayala et al. (2009) concluded that treatments were largely equivalent regardless of the type of intervention (e.g., exposure, applied tension), with 70–80 % of patients experiencing clinically significant change. Despite expectations, limited evidence was found for the additional effects of applied tension beyond exposure alone when they were combined in the same treatment. While the treatment research with adults is favourable in outcome for BII, as noted above, BII phobia has largely been neglected in the child and adolescent treatment literature. Furthermore, while OST has been deemed to be *well-established* for the treatment of childhood-specific phobia in general, there is no efficacy data for its use with BII phobic in youth.

OST Approach for BII Phobic Youth

It is evident from the adult literature that this phobia type has a complex clinical presentation involving a unique emotional (e.g., fear, pain and disgust) and physiological (e.g., fainting) response. The clinical phenomenology of BII in youth has not yet been fully explored; however, if consistent with adults this multifaceted presentation needs to be considered in the delivery of treatment. In youth, the clinical presentation may be further complicated by parent psychopathology as BII has one of the strongest heritability estimates of any phobia type (Öst 1992; Van Houtem et al. 2013). Moreover, BII phobic youth may be more difficult to engage in exposure therapy. They may have trouble differentiating the therapist from other health professionals (e.g., doctor or nurse) who they associate with past anxiety-provoking experiences as suggested by Öst et al. (2001). Furthermore, BII phobic youth may be less motivated than other phobia types to overcome their fear, and as such, it may be more challenging to establish a trusting relationship with these youth, which is critical in terms of facilitating exposure. Implementing an OST approach for BII phobic youth requires exposure to blood and medical procedures and thus the involvement of qualified health professionals, such as nurses or doctors in addition to the child's treating therapist, which can complicate the feasibility and delivery of exposure in routine clinical practice (see Ollendick et al. 2009).

Treatment Illustrations

The following case studies describe the treatment response of two children to OST for BII phobia. The first case follows a standard OST format (e.g., OST without specific adaptations and without a formal maintenance program). Following this, the second case involves a formulation-driven approach to delivering OST with adaptations to the traditional OST structure in order to address BII-specific issues and maximise outcomes (refer to Fig. 1).

Case Study 1

Louise (not her real name) an 8-year-old girl presented with a specific phobia of injections. Louise's mother (Kathryn) reported that Louise would become highly anxious prior to going to the doctor or dentist for fear of having an injection. Recently, Louise's brother had required a whooping cough vaccination. In the days leading up to her brother's appointment, Louise constantly asked for reassurance that she would not be vaccinated also and had extreme trouble sleeping. While at her brother's appointment (her mother required her to go) Louise turned away,

closed her eyes and blocked her ears when he was being vaccinated. She also reported that in the past the family had planned to travel to Bali, Indonesia for a holiday; however, Louise had refused the necessary vaccinations. In addition to her fear of injections, Louise was also concerned about being injured. Louise refused to ride her bike for fear of falling off and getting hurt. In the past, she had been hit in the face by a netball and following this incident had refused to play the sport again, resulting in the family withdrawing her from the team.

Kathryn reported that when Louise was 5 years old she had taken her for a blood test and that the nurse had "botched it". Kathryn recalled that Louise had an extreme reaction crying, kicking, and screaming, which resulted in her having to be restrained. After the nurse had inserted the needle, she had been unable to find Louise's vein. Following the first failed attempt, the nurse was going to attempt a second blood draw; however, Kathryn refused and had taken Louise home. Kathryn returned to the doctor who said they could pursue a less invasive assessment approach that would not involve a blood test given Louise's distress. Louise had not had any injections since that time.

Assessment

Louise and her mother were interviewed using the parent and child versions of the Anxiety Disorder Interview Schedule for Children (Silverman and Albano 1996). Based on Louise's report and that of her mother during the diagnostic interviews, Louise was diagnosed with specific phobia of BII with a clinician severity rating (CSR) of 6, indicating a moderately severe fear. Additionally, Louise was diagnosed with a number of comorbid conditions including generalised anxiety disorder (GAD; CSR 6), separation anxiety disorder (SAD; CSR 5), a specific phobia of the dark (CSR 5), and a specific phobia of vomiting (CSR 4). During the initial assessment, Louise was also administered a BAT. The BAT required Louise to enter a room with a nurse, sit on a chair, and be prepared for a blood test (e.g., tourniquet placed on her arm, alcohol wipe, nurse pretends to draw blood). Louise refused to enter the room and rated her subjective anxiety at 7 (on a scale ranging from 0 to 8).

During the functional assessment of her phobic beliefs, Louise stated that her greatest fear was the pain associated with getting an injection. She reported that she was afraid the nurse would force her to have the injection and that the nurse would be careless and insert the needle into the wrong part of her body. She also reported a heightened sense of disgust, describing injections as disgusting because they "break through the skin and veins which is gross". The therapist and Louise constructed a hierarchy of

injection-related stimuli. At the end of this session, Louise's therapist discussed with the family the rationale for OST approach.

Formulation

Louise's temperament in combination with negative learning experiences may have led to the development of her injection and injury phobia. Louise was reported to be a "difficult" baby, crying for long periods of time and difficult to settle. The complicated blood test taken when Louise was 5 years old and following this her mother's response to the situation (e.g., refusing to allow the nurse to attempt a second blood draw and facilitating escape from the clinic) may have reinforced Louise's fear of injections and lack of trust for health professionals. Moreover, in the years following this incident the family had discussed this experience in a highly negative manner with both Louise and Kathryn reporting that the nurse had "botched it" and that she was "unfriendly and rushed" thereby perpetuating a fear and distrust of medical professionals.

Consistent with the proposed CBT formulation of BII phobia in youth (see Fig. 1), Louise had numerous catastrophic phobic beliefs associated with pain that were maintaining her fear of injections and injury. More specifically, Louise reported exaggerated danger expectancies whereby she believed that the pain of having an injection would be unbearable and furthermore, underestimated her ability to cope with experiencing such pain (e.g., needles or falling off her bike). Additionally, she believed that health professionals (e.g., nurse, doctors) were careless and would insert the needle into the wrong part of her body. She also believed she would be forced to have an injection. Louise's fear was maintained by her avoidance of stimuli that she associated with pain, such as injections and situations during which she had previously been hurt (e.g., riding her bike and playing netball). In the short term, Louise's avoidance reduced her distress; however, this avoidance prevented her from gaining corrective evidence that she could indeed tolerate pain and cope with the challenges of the task, and that the majority of health professionals were careful, helpful and could be trusted. Louise also demonstrated attentional biases whereby in the presence of BII stimuli she became extremely avoidant (i.e., when her brother had a vaccination; refer Fig. 1, CBT model). Attentional avoidance perpetuated her fear and phobic beliefs and prevented the acquisition of safety learning.

Louise's parents facilitated and accommodated her avoidance and provided her with undue attention when she engaged in anxious behaviours, therefore inadvertently rewarding her anxious avoidant responding to challenging situations. Louise's parents changed their holiday plans so

as not to travel to a country that required vaccinations. They additionally allowed her to leave her netball team and did not encourage her to continue riding her bike. Additionally, Kathryn reported feeling highly anxious when her children were being vaccinated and hence it is likely that Kathryn modelled anxious behaviours to Louise when her children were being vaccinated. Certainly Kathryn also provided negative verbal information about needles and health professionals. In accordance with the proposed CBT model of BII phobia in youth (refer Fig. 1), these parenting factors were hypothesised to perpetuate Louise's fear.

Treatment

Louise's single-session treatment followed the standard format of OST (3 h of graduated exposure, cognitive challenges, modelling, reinforcement of approach behaviours, and psychoeducation) given that to date there was no clear evidence regarding the effectiveness of this treatment for BII phobic youth (see Table 1 for a summary of the session). The first hour of treatment was conducted at the Griffith Psychology Clinic and commenced with the therapist providing Louise with psychoeducation about needles and showing her pictures of people having injections. Following this, the therapist and Louise watched videos of people having blood tests. While watching the video Louise indicated that she thought it was disgusting and "so gross" seeing people have a blood test. Through the use of participant modelling, Louise and the therapist held a syringe with a capped needle and administered a pretend blood test to Kathryn. The session progressed with Louise having a tourniquet placed on her arm and finally the therapist giving her a pretend blood test. Louise was positively reinforced with attention and praise when she throughout the session when looking at and holding the BII stimuli.

The second and third hours of treatment were conducted at the nearby Griffith University Medical Centre. Louise and the therapist met with a pathology nurse. The nurse provided Louise with education regarding the benefits of having injections and also spoke to her about her job (e.g., taking 20+ blood tests per day to people of all ages including new born babies). Exposure was conducted in the form of behavioural experiments. Each time the therapist would ask Louise what she feared would happen, and then following this, they discussed whether her prediction was correct. The therapist and Louise had the nurse give them pretend blood tests, as well as Louise holding an uncapped needle. Louise watched the therapist model having two real blood tests. Prior to observing the therapist, Louise reported her fear to be a 5 (on scale ranging from 0 to 8), following completion of the second blood test she indicated that her fear had reduced to a 1. At the end of the second

Table 1 Summary of Louise's OST session

OST	Exposure steps	Fear rating pre (0–8)	Fear rating post (0–8)
Hour 1	Psychoeducation regarding needle-related stimuli (e.g., syringe and tourniquet)	1	0
	Looking at pictures of needles and people having injections/blood tests	1	0
	Videos of needles	4	1
	Practise the steps involved in having a blood test	1	0
Hour 2	Psychoeducation regarding the benefits of having injections and blood tests and the role of a pathology nurse	0	0
	Pathology nurse prepare the therapist for a blood test	0	0
	Pathology nurse prepare Louise for a blood test	3	0
	Pathology nurse holding an uncapped needle above Louise's vein	6	1
	Observe the therapist having two blood tests	4	1
	Pathology nurse complete a blood test in a training skin pad attached to Louise's arm	5	0
	Observe the pathology nurse and therapist have a finger prick	2	0
Hour 3	Louise give the training skin pad and therapist a finger prick	2	0
	Louise rest the finger prick on her finger without releasing it	4	1
	The pathology nurse hold the finger prick on Louise's finger without her releasing it	7	3

hour, Louise was encouraged to have a blood test, however she refused.

Since she was unable to have the blood test following the second hour of treatment, the third hour focused on finger pricks. Louise observed both the nurse and therapist have a finger prick. She practised putting finger pricks into the fake skin on the training pad and also administered them to the therapist. Participant modelling was also used whereby Louise placed her finger underneath the therapists while they had a finger prick. Following this, Louise practised resting the finger prick (without pressing the release button) on her own hand. Shortly after this, to elicit Louise's phobic beliefs regarding health practitioners, the therapist suggested that the nurse hold the finger prick against Louise's finger. Louise refused and became upset crossing her arms, scowling, and had tears in her eyes. She stated that "you said I would not have to do anything I did not want to". Louise reported her fear level to be a 6 on scale ranging from 0 to 8. The therapist asked Louise what she feared would happen. She indicated that she worried the nurse would surprise her and give her a finger prick. The therapist assisted Louise to test this belief using a behavioural experiment. Subsequently, Louise allowed the nurse to rest the lancet on her finger for 10 min. She rated her fear at a 5 (on scale ranging from 0 to 8) when commencing the task with this gradually declining to a 1 at the conclusion of the 10 min. The therapist discussed with Louise as to whether her belief had been correct. Louise was able to provide a more correct and adaptive appraisal of health professionals stating that "doctors and nurses can

be trusted and if they do give me an injection it is to help me stay healthy and well".

Throughout the session the therapist frequently praised Louise for facing and coping with her fears. At the conclusion of the session, Louise was comfortable interacting with the nurse and needle-related stimuli (e.g., holding needles, giving finger pricks); however, she still had not had a needle penetrate her skin. Louise and her parents were reminded that this session was just the beginning of Louise's treatment and that she would need to continue with exposure practice outside of therapy for several months.

Post-Treatment and Follow-Up Assessment

One week after her OST, Louise's fear had reduced from her pre-treatment assessment. On the ADIS-IV-C/P, Louise continued to meet criteria for a diagnosis of BII phobia; however, her CSR had reduced from 6 to 5. She also continued to meet criteria for specific phobias of the dark (CSR 5) and vomit (CSR 4). One month later, Louise was reevaluated. Louise's parents reported that they had been attempting finger pricks and practicing pretend blood tests at home, but had made limited progress. Louise still had not had a finger prick. However, Kathryn indicated that they were applying the strategies they had learnt during Louise's OST to her fear of the dark with great success. Louise had been sleeping in her own room and had only a smaller lamp turned on while sleeping. Louise's ADIS-IV-C/P diagnosis at this time was a specific phobia of BII

(CSR 5). During a BAT (identical to that completed at pre-treatment), Louise was able to enter a room and be prepared for a blood test demonstrating improvement in her ability to approach and attend to the feared stimuli. However, her subjective anxiety was at a 7 (scale of 0–8) and she was not able to have the blood test.

Complicating Factors

There were a number of challenges that arose in delivering standard OST to Louise. Throughout the treatment process, Louise seemingly lacked motivation to overcome her fear of needles. Upon arrival at all sessions, she would state that she would not be having any injections. Typically tangible reinforcers (e.g., food, toys, stickers) are not used during OST with social reinforcers favoured (Rudy and Davis III 2012); however, in Louise's case social reinforcers (positive attention, praise and social support) had a limited effect. Establishing trust with Louise was another challenge. Louise feared health professionals would force her to have an injection, which led to her extreme control and refusal during session to engage in certain tasks. Prior to her OST, only two brief assessment sessions were carried out with Louise, both involving semi structured interviews. Although the therapist was able to successfully build rapport with Louise, given her past negative experiences with medical professionals she was hesitant to engage with the therapist during exposure due to a fear of forcibly being given an injection. Throughout her OST, Louise repeated on multiple occasions "you told me I would not be made to do anything I don't want to do and I am not going to have an injection today". Particularly, during the third hour of treatment, exposure came to an impasse in that Louise had completed all possible easier exposure steps (e.g., resting a finger prick against your skin) and was at the point that to move forward she would have had to have a needle penetrate her skin. The therapist was unable to move past this point in the session either with finger pricks or blood tests. Because of this, Louise's catastrophic cognitions relating to pain remained unchallenged. Following OST, Louise and her mother occasionally engaged in exposure practice at home; however, when practising they always completed the same exposure step (e.g., pretend blood test) which Louise was able to carry out quite easily as she did not have a needle penetrate her skin. The family did not actively seek opportunities for more challenging exposures, such as scheduling appointments with a doctor or visiting a pathology clinic. Given that the ultimate goal of OST is to assist children in facing their fears (i.e., having an injection in this case), this approach was serving to perpetuate Louise's fear that medical professionals will push you into having a procedure you are terrified of. We hypothesised that Louise's poor progress was a function of insufficient

preparation and her session being too fast for her. Louise would have benefited from more exposure to challenge her beliefs that all medical professionals are untrustworthy and will force you to have injections.

Case Study 2

Ivy (not her real name) an 11-year-old girl presented with a specific phobia of injections. Ivy's father was quadriplegic and if he caught a cold or flu he could develop pneumonia and become very ill. Because of this, her father's doctors recommended that Ivy and her mother have a flu vaccination every year. Due to her injection phobia, Ivy had not had a flu vaccination for the past 2 years. Ivy previously had all her early childhood vaccinations; however, Rosie (Ivy's mother) indicated that when being vaccinated Ivy had to be restrained and that she would cry, scream, punch, and hit in an attempt to escape the doctor. At age 9, Ivy had required a blood test. She reported a range of physiological symptoms (e.g., dizziness, pale, cold sweat, racing heart, and feeling faint) during the test and immediately following its completion had fainted. Ivy was reported to be able to watch medical shows on television; however, she was unable to watch others receive injections in real life. Rosie reported that her husband required regular injections due to his illness and that she administered these at home. This often distressed Ivy who had to leave the room when this was occurring. She also asked her mother to keep her father's needles out of sight. Ivy stated that her greatest fear was having an injection in her mouth while at the dentist. When Ivy had a wobbly tooth she would refuse to eat solid food and would only consume a liquid diet. In addition to a fear of injections, Ivy worried about accidentally swallowing her loose teeth.

Assessment

Ivy and her mother were administered the ADIS-IV-P and ADIS-IV-C (Silverman and Albano 1996). Ivy received a principal diagnosis of a specific phobia of BII (CSR = 7) and comorbid diagnoses of a specific phobia of dentists (CSR = 5) and specific phobia of dogs (CSR = 4). Ivy completed a BAT as part of her initial assessment. Consistent with Louise, the BAT involved Ivy being prepared for a blood test. Ivy was able to complete all steps involved in being prepared for a blood test; however, she rated her subjective anxiety at a 7 (on scale ranging from 0 to 8).

The therapist completed a functional assessment of Ivy's phobic beliefs. Ivy reported that she worried that the needle would hurt so much that she would prefer to die. She also indicated that she was concerned that the health professional would inject the wrong medicine into her and that this would kill her. Furthermore, she stated that her gums

were harder than her arm and therefore a dental injection would hurt considerably more than an injection in the arm. She also reported worrying that if she had blood test she would faint.

Formulation

Ivy was reported to have had a number of specific fears (e.g., dogs, butterflies) throughout her early childhood. Her mother, Rosie, indicated that she herself was afraid of injections for many years; however, she overcame her fear following her husband's accident, which resulted in her needing to give him daily injections. Also, members of Rosie's extended family had a history of fainting when exposed to blood and injection-related stimuli. Given her family history and possible anxious temperament, it is probable that Ivy had a biological predisposition to developing her own injection phobia (refer Fig. 1). Ivy also had multiple negative learning experiences associated with injections. During her most recent blood test, Ivy experienced a number of aversive physical symptoms, which may have led to the development of a number of dysfunctional beliefs. Consistent with the proposed CBT model of BII phobia in youth, Ivy had exaggerated beliefs regarding the severity of pain, as well as the likelihood of something bad happening (e.g., fainting, dying). Moreover, Ivy underestimated her ability to cope with needles and injections to the point that she believed death would be preferable to enduring the pain. In addition to experiencing fear, Ivy also reported experiencing intense disgust associated with injections, and particularly those in the mouth (see Fig. 1). It was hypothesised that disgust concurrently interacted with Ivy's fear and led to greater avoidance of injection-related stimuli. Moreover, Ivy's family appeared to accommodate her avoidance of injections. It was hypothesised that Ivy's avoidance and her parent's accommodation of this maintained her fear, as she has been unable to acquire corrective evidence that she can tolerate and cope with the pain, disgust, and physiological symptoms associated with fainting.

Modifications to Standard OST

Given the challenges described in delivering standard OST to Louise and Ivy's individualised cognitive behavioural formulation, a number of modifications were made to Ivy's treatment to maximise her outcome. Ivy was provided with education regarding the relationship between pain and fear and coping with discomfort. Moreover, she learnt about arousal and the experience of fainting. Ivy was also given education to normalise her physical response to BII stimuli and was taught strategies to assist her to tolerate the unpleasant physical symptoms she experienced when

having an injection or blood test. A more incremental approach to exposure was taken with Ivy using a topical anaesthetic and dry needling (e.g., acupuncture) prior to exposure to blood tests. In addition to fear, when exposed to BII stimuli, Ivy responded with disgust and because of this a number of the exposure tasks she completed were designed to elicit this emotion. To motivate and increase Ivy's engagement during the OST, tangible rewards were used throughout her session. To enhance compliance with exposure practice following OST, an education session and an e-therapy maintenance programme were developed and included as part of Ivy's treatment. Furthermore, Rosie directly observed Ivy's OST session and participated in all e-therapy maintenance sessions. It was evident from Louise's treatment that parent involvement would be particularly necessary for BII phobic youth due to difficulties in organising post-treatment exposure opportunities, parent's accommodation of their child's phobic avoidance, and the parent's own BII fears. The inclusion of the education session also provided another opportunity for the therapist to gain Ivy's trust. Below is a detailed description of Ivy's modified treatment approach.

Parent and Child Education Session

One week prior to their OST session, Ivy and Rosie attended a 1-h education session. The session focused on education about the aetiology of BII phobia, the phobic response (cognitive, physiological, and behavioural), an overview of exposure therapy, the role of parenting practices and family accommodation in the maintenance of a phobia, parent contingency management training, and a rationale for OST. During the session, the therapist worked with Ivy and Rosie to develop an exposure hierarchy (e.g., starting with videos of people having dental injections and progressing to having a blood test) to help guide the OST. The therapist, Rosie, and Ivy also spent time planning and preparing for Ivy's exposure practice following OST (e.g., purchase syringes from the local chemist, book doctor and physiotherapist appointments). The therapist emphasised to the family that the OST was just a "kick start" and that a fear Ivy had for long time would not disappear after one session. The therapist advised the family that the e-therapy maintenance sessions were still an active part of treatment and that if Ivy continued to practise exposure tasks regularly, with the support of the therapist, over the weeks following her OST, she would be able to overcome her fear.

Treatment

Ivy's OST involved 3-h graduated exposure in addition to the use of cognitive challenges, modelling, and

reinforcement of approach behaviours (see Table 2 for a summary of the session). The first hour commenced at the Griffith University psychology clinic. Rosie was involved throughout the session observing and also as a model demonstrating particular exposure steps (e.g., having a finger prick). The session began with reviewing the psychoeducation material regarding the phobic response. Ivy was asked to identify the physiological cues she experienced when having an injection. She was encouraged to be aware of these cues throughout the session and if she noticed them to try a number of adaptive responses to manage these symptoms (e.g., wiggle her toes, take some slow breaths, have a drink, splash water on her face, sit or lie with her feet up).

Ivy and Rosie were also provided with education about pain (e.g., pain cells send a message to the brain) and the interaction between pain, anxiety, and disgust. Throughout the session, a narrative/language associated with pain experiences was used to challenge Ivy's dysfunctional phobic beliefs. The therapist encouraged Ivy to use less intense language to describe her pain for example using the terms "hurt" and "discomfort" to highlight to Ivy that we experience different levels of discomfort and that the pain associated with an injection is transient and manageable, as opposed to a chronic and enduring pain that is associated with considerable suffering. Ivy was taught that anxiety tricks you into believing that the pain associated with an

injection is unbearable; however, in reality pain is part of our everyday lives and although it is unpleasant we frequently tolerate and cope with pain. The concept of an "ouch" thermometer was introduced to Ivy, to rate her level of discomfort from (0 to 8). A discussion was had with Ivy and her mother about painful experiences that Ivy had been able to cope with in the past (e.g., spraining her arm playing hockey, biting her tongue, stubbing her toe and falling off her bike). Ivy was asked to rate the discomfort she experienced during these events on the ouch thermometer (e.g., spraining her ankle—6/8) and then to compare this to the discomfort associated with having an injection. A behavioural experiment was then carried out whereby Ivy and her mother were both asked to hold ice for as long as they could. After 30 s of holding the ice, Ivy reported that her hand was hurting (she rated 5/8 on the ouch thermometer); however, she said she knew she could keep going and persisted with the task for 8 min. Following the task, Ivy was questioned as to how she had been able to cope with the discomfort. She was able to reappraise the pain associated with holding ice and stated "I felt uncomfortable while holding the ice, it hurt, however I knew I was not in danger and could handle it". Ivy was informed that when someone is anxious it turns up the pain dial in their head (e.g., more painful); however, if they stay calm and use calm thoughts it is much easier to cope with discomfort.

Table 2 Summary of Ivy's OST session

OST	Exposure steps	Fear rating pre (0–8)	Fear rating post (0–8)
Hour 1	Psychoeducation regarding coping with pain and disgust	0	0
	Exposure task—holding ice cubes	0	0
	Watching videos of children receiving dental injections	3	1
	Reward = Points on behavioural chart		
Hour 2	Ivy holding a small dry needle	1	0
	Watching the therapist and her mother receive a small dry needle	1	0
	Ivy (under the supervision of the physiotherapist) tap a small dry needle into her own arm	3	0
	Physiotherapist tap a small dry needle into Ivy's forearm which had topical anaesthetic on it	4	1
	Having a small dry needle in her forearm that had no topical anaesthetic	5	1
	Having a large dry needle in her forearm with topical anaesthetic	2	0
	Reward = Iced chocolate from university coffee shop		
Hour 3	Sitting in a dental chair and watching a dental injection video	3	1
	Wobbling her loose tooth while sitting in the dental chair	2	1
	Watching the therapist have a finger prick	0	0
	Giving the therapist a finger prick	0	0
	Watching the therapist have a blood test	5	0
	Having a pathology nurse pretend to give her blood test with an uncapped needle	4	0
	Having a blood test in her arm, which has topical anaesthetic	7	2
	Reward = \$10 iTunes gift card and candy blood bag		

The remaining time during the first hour was spent watching dental injection videos aimed at eliciting disgust, in order to both normalise the disgust response, as well as reduce the intensity of this emotion through exposure. Videos were chosen specifically to elicit Ivy's disgust (as opposed to fear), given that Ivy did not perceive herself to be in imminent danger watching a video. Exposure was conducted in the form of behavioural experiments with Ivy predicting that she would find the videos disgusting and that the child in the video would scream out in pain. Ivy watched the video multiple times, and at the end of the task her predictions were reviewed. Ivy reported that she initially felt gross while watching the video, but over time she got used to it. She also indicated that the child in the video did not appear to be in pain. The therapist reinforced to Ivy that disgust is another normal emotion that is unpleasant, but that we can cope with it and, like fear, passes with time if we stay in the situation and allow ourselves to face it. At the beginning of the session, the therapist introduced the concept of a scoreboard to Ivy for "braving discomfort and fear". Ivy received points for braving and coping with discomfort and fear throughout the session. At the end of the second hour and third hours, Ivy was given rewards for achieving a pre-determined number of points.

The second hour of Ivy's OST was conducted at the Griffith University Physiotherapy Clinic. During this hour, a physiotherapist, trained in dry needling, (i.e., acupuncture) gave Ivy multiple dry needles. Dry needling uses very fine needles of varying sizes and thicknesses providing an ideal opportunity to graduate exposure to injection needles. To assist with graduating Ivy's exposure tasks, a topical anaesthetic cream was placed on her arm. At the conclusion of the hour, Ivy was able to have more than 15 dry needles of various sizes initially in her arm with the anaesthetic, and then in her non-anesthetised arm. Ivy initially worried that the pain associated with the dry needles would be unbearable. Over the course of the hour, however, she reappraised the level of discomfort associated with having a dry needling saying to Rosie "the dry needles sting however only for a short time and they do not hurt as much as when I broke my finger at netball, which I was able to cope with".

The third hour was initially conducted at the Griffith University dental clinic and then moved to the medical centre where the therapist and Ivy met with a pathology nurse. At the dental clinic, Ivy sat in a dentist chair, looked at dental tools, and practised wobbling her teeth for extended periods of time. Ivy's predictions regarding the fear and discomfort she would experience when completing exposure tasks declined during her time at the clinic, and she was able to successfully reappraise beliefs related to her ability to cope with discomfort and disgust.

Following this, the session progressed to the medical centre where Ivy watched the therapist have a blood test. While watching the therapist Ivy became very pale and reported feeling nauseous. The therapist encouraged Ivy to take some slow breaths. She then had her sit in a chair with her feet up and fan her face to cool herself down. Ivy stated that she was worried she would faint. The therapist used this opportunity to conduct some challenges of her phobic beliefs—What are the chances you will faint? How bad would it be? If you did faint would you be able to cope with that? Ivy was encouraged to use cognitive challenges, such as "I feel icky but I am ok—the feelings will pass" and "this is just a trick—I am not in danger". Once the colour had returned to Ivy's face, the exposure task was continued with the therapist having a second blood test. Ivy said she felt "yucky" but she was not in danger watching someone have a blood test and the feelings passed. Towards the end of the session, Ivy was able to have a blood test in her arm that had a topical anaesthetic on it. She did not cry and was not restrained at all. Following the blood test, she reported her actual pain level to be significantly lower than she had anticipated. She reported that the injection had hurt a little; however, she was able to cope with the pain. At the end of the session, Ivy's progress was reviewed and the family was reminded that the session was a "kick start" in overcoming her fear. A time was scheduled to provide Ivy with e-therapy maintenance sessions to assist the family with ongoing exposure practice, problem solve any obstacles to exposure and provide ongoing motivation and reinforcement to both Ivy and her mother.

Maintenance Program

Following Ivy's OST, the clinician contacted Ivy and Rosie once a week for a 30-min e-therapy session (delivered over Skype) for 4 weeks. During the sessions, the therapist reviewed Ivy's progress, completed a brief exposure task, and then helped plan Ivy's exposure practice for the following week. Exposure tasks completed by Ivy over the course of the 4 weeks included visiting an acupuncturist, wobbling her loose tooth, visiting a dentist and sitting in the chair, having her teeth cleaned by the dentist, giving her mother a finger prick and having her own finger prick. In addition to her injection-related fears, Ivy and her mother were encouraged to apply the skills they had learnt to her fear of dogs. Ivy and her mother developed their own exposure exercises with the therapist's assistance and visited a friend's house that owned dog and went walking in a dog park. Rosie regularly rewarded Ivy (e.g., ice cream, friend sleep over) for practising her exposure tasks.

Post-Treatment and Follow-Up Assessment

A post-assessment was carried out 1 week after Ivy's OST session. On the basis of the ADIS-IV-P/C, Ivy continued to meet criteria for BII phobia; however, the severity of her phobia (CSR = 4) had reduced significantly. Although she continued to meet criteria for a dental phobia (CSR = 4), her dog phobia diagnosis was now considered subclinical (CSR = 3). At 1-month follow-up, Ivy did not meet criteria for any diagnosis on the ADIS-IV P/C. On the BAT task (administered at 1-month follow-up), Ivy was able to be prepared for a blood test and reported minimal levels of distress completing the task (1 on a scale ranging from 0 to 8).

Modified OST for BII Phobia

The aforementioned case studies highlight some of the unique challenges associated with treating BII phobia in children and adolescents. BII phobia has a highly heterogeneous presentation in youth. A formulation-driven approach is essential to the delivery of an effective treatment for these children and adolescents. The following is a summary of key maintaining mechanisms (see Fig. 1) that need to be considered when treating these youth and a discussion of possible modifications to a traditional OST approach to augment treatment outcome. Specifically, our treatment approach includes modifications to address (a) pain, (b) disgust, (c) the physical symptoms associated with BII (e.g., fainting and vomiting), (d) parent involvement in treatment, and (e) maintenance of treatment gains (Table 3).

A key treatment consideration for BII phobic youth is addressing the role of pain in the maintenance of their fear. Previous research with adults has found that BII and dental fearful individuals have a tendency to exaggerate or overestimate pain and have a lower perceived ability to tolerate physical pain and discomfort (Arntz et al. 1990; Severeijns et al. 2001; Smith and Meuret 2012). For children who report catastrophic cognitions and phobic beliefs associated with pain, it is suggested that their psychoeducation focus on the interaction between pain and fear. Addressing phobic beliefs associated with pain occurs through education about the experience of pain (e.g., pain cells) and by changing the language children and parents use to describe pain; the use of an ouch thermometer demonstrates to youth the various intensities of discomfort a person may experience from slight discomfort to extreme discomfort; behavioural experiments test exaggerated danger expectancies (probability and severity) related to pain and discomfort, as well as coping estimates; and explicitly

having children reappraise their phobic beliefs using knowledge obtained via adaptive learning experiences.

Öst et al. (1997), in their description of OST for adults with injection phobias, report that their patients had been able to have multiple finger pricks, saline injections, and multiple blood draws by the end of the session. From the above case studies, it is evident that the pace of exposure for BII phobic children is significantly slower. To address this dry needling (i.e., acupuncture) and a topical anaesthetic are recommended as intermediate exposure steps prior to the finger pricks and blood tests. Dry needles do not look like traditional needles as they are smaller, finer, and are not attached to a syringe. Children may be more willing to attempt this during exposure. These transitional exposure steps give children the opportunity to have a corrective learning experience. They learn they can cope with discomfort and once children have had one needle they are often more willing and open to engaging in other exposure tasks. Moreover, after initially attempting exposure tasks on their arm with the topical anaesthetic, children are encouraged to then attempt needles in their arm without anaesthetic with the goal to ultimately phase out the use of the anaesthetic.

Furthermore, the use of a reward system during the OST session is considered integral in engaging children and keeping them motivated. With Ivy a scoreboard was used, receiving points for braving her discomfort and fear. Points are awarded throughout the session, and at the end of the second hour and third hour, rewards are presented for achieving a pre-determined number of points. Hence, contrary to traditional OST, tangible rewards may be necessary to encourage youth who appear to be more difficult to engage in exposure therapy. Similarly, Rudy and Davis III (2012) recommend the use of tangible rewards during OST for special populations, such as children with Autistic Spectrum Disorders or Intellectual disability, who are more likely to have low motivation. During the education session, contingency management strategies are also introduced to parents and a rewards programme set up at home to keep children motivated to continue exposure practice following treatment.

Many BII phobic youth report previous experiences of being restrained or forced to have an injection and hence they often have limited trust of their therapist and the other clinicians involved in their treatment. This serves to further heighten their anxiety during the OST session. For youth who have comorbid chronic illnesses and BII phobia, this may particularly be the case if they have required life-saving treatments. When conducting exposure it is important that the child feels in control of the session. Children should never be forced to have an injection during the session. Wherever possible it is helpful to give the child a

Table 3 Suggested adaptations to treatment for BII phobia in children and adolescents

Characteristic of BII phobia	Mechanisms proposed in CBT formulation	Adaptation for formulation-driven OST
Pain	Pain sensitivity Direct/classical conditioning Overestimation of threat Underestimation of coping Avoidance	Psychoeducation regarding the role of pain and practice coping with discomfort Exposure to discomfort/pain (i.e., holding ice, pulling off band aids, pulling out a piece of hair) Reappraisals about expectancies of pain (probability and severity) and estimates of coping—using an “ouch! Thermometer” Exposure tasks: Dry needling (e.g., acupuncture), finger pricks, use of topical anaesthetic for blood test
Disgust	Disgust sensitivity Appraisals of disgust Physiological symptoms (e.g. nausea) Avoidance	Psychoeducation regarding tolerating disgust Disgust evoking exposure tasks Reappraisal of expectancies of disgust (probability and severity) and estimates of coping
Physiological symptoms	Overestimation of threat Underestimation of coping Negatively reinforced avoidance behaviour through reduced symptoms of faintness, nausea and embarrassment	Psychoeducation regarding fainting Encouraging the use of adaptive strategies to cope with physical symptoms—e.g., lie down, drink cool water, wiggle toes etc. Reappraisals about expectancies associated with unpleasant physiological symptoms (probability and severity) and estimates of coping
Parent BII-related fears	Vicarious conditioning Negative information transmission Parent psychopathology Parenting practices such as positive reinforcement of avoidance	Parent education session prior to OST—contingency management training Observation of the therapist modelling how to manage child anxiety Involvement in ongoing maintenance—e-therapy
Lack of motivation	Parenting practices such as lack of positive reinforcement for approach behaviours	Tangible rewards Contingency management training with parents
Difficulties maintaining treatment gains	Parenting practices—familial and contextual obstacles to homework compliance Avoidance	Education session—discuss preparing for exposure and homework E-therapy maintenance program/phone calls over 1 month following OST
Lack of perceived control/ lack of trust for health professionals	Direct/classical conditioning Overestimation of threat Underestimation of coping	Additional time spent with the child during education session building rapport Slow pacing of OST Child-led exposure tasks with child choosing next exposure step Reappraisal of cognitive biases associated with health professionals being untrustworthy (e.g., It is the priority of health professionals to keep us safe and well. This may involve procedures that make us uncomfortable, but are intended to keep us safe and well)

choice so that they feel more in control (e.g., *Would you like the nurse to give you a finger prick or would you prefer to give one to yourself?* or *Would you like the physiotherapist to tap in your dry needle or would you like the physiotherapist to set it up in the right position and then you tap it yourself?*). It is also important when discussing the rationale for treatment to emphasise to the child that they will be in charge of how the session progresses.

Additionally, if a child was becoming overwhelmed or finding an exposure task to difficult the therapist can have the child take a step back and complete an easier exposure task in some circumstances. For example, when a child was attempting a finger prick, the therapist may leave the room for a short period and allow the child to attempt the task without the pressure of therapist, parent, or other health professionals observing.

Another unique challenge associated with BII phobia and which needs to be considered when delivering treatment to youth with BII is the role of disgust in the aetiology and maintenance of the disorder. Louise and Ivy reported that they found BII-related stimuli disgusting. With Ivy, disgust was introduced when explaining to the family the phobic response (cognitive, physiological and behavioural) during the education session. In the first hour of OST, children can be taught that disgust like pain and anxiety is another normal feeling that we can learn to tolerate. A distinction is made between fear that occurs in response to danger, versus anxiety or disgust that occurs in response to something that is not dangerous. Children are educated that they need to learn to turn the dial down on anxiety and disgust because they do not have to avoid stimuli, which is not dangerous. Moreover, in accordance with the suggestion by Olantunji et al. (2007a, b), it is recommended that the therapist complete disgust evoking exposure tasks with the child similar to those used with Ivy in order to assist in habituation of disgust as well as the development of adaptive and modest appraisals of disgust. For example, the therapist could squeeze their finger after having a finger prick so that blood flows or drips from it or the therapist and child could look at pictures of infected wounds. Targeting disgust is proposed to lead to enhanced treatment outcome; however, further research is needed to clearly establish this with BII phobic youth.

An important feature distinguishing BII phobia from all other phobia types is its unique physiological response. In the cases reviewed, Ivy had a history of fainting when confronted with her phobic stimuli, which is a common feature of BII. Applied tension, a coping technique that is implemented during exposure to BII stimuli with adults, was developed to counteract the unique vasovagal fainting response experienced by individuals with BII phobia. However, in a recent review, Ayala et al. (2009) concluded that the findings in relation to the superiority of applied tension above and beyond exposure alone were mixed, with little compelling evidence for the need of applied tension. Hence, exposure alone was used to treat Ivy. For BII phobic children who faint or vomit, it is recommended that the OST commence by normalising this physiological response and educating children about the evolutionary purpose of fainting/vomiting when exposed to blood (Antony and Watling 2006). The child is encouraged throughout the session to be aware of their bodily cues and if they notice that they are starting to feel faint they are reminded to stay calm and to use adaptive strategies (e.g., lie down, drink cool water, wiggle toes) to manage these normal symptoms, which do not present any danger or risk. Once they begin to feel better, children should continue with exposure as planned.

The case studies presented also highlight a number of parent-related issues that may interfere with maintenance of children's treatment gains following OST. As was evident in both cases, parents reported that they too were fearful of BII-related stimuli and Ivy had a family history of fainting. If parents are phobic themselves, they may avoid assisting their child with exposure tasks and model anxious behaviours when interacting with BII stimuli. An integral part of Ivy's treatment was the education session during which her mother was provided with information regarding the aetiology of phobias (e.g., heredity, learning pathways) and strategies to facilitate Ivy's exposure such as being aware of verbal messages and modelling. Moreover, having parents actively involved in their child's OST is also important as it gives them an opportunity to vicariously expose themselves to BII stimuli. Parents of children with phobias often facilitate their child's avoidance of phobia-related objects and situations (Milliner et al. 2013) and thus inadvertently reinforce their anxious behaviour. Including the education session in addition to OST provides an opportunity to teach parents contingency management strategies such as using attention and rewards to reinforce approach behaviours. Having parents involved throughout the session also gives them an opportunity to observe the therapist respond appropriately to their child's anxious behaviour.

Relative to other phobia types, such as dog or dark phobias, organising exposure practice for BII phobia takes significantly more time and active planning. Families may need to schedule doctor's appointments for vaccinations, take their child to see physiotherapists for dry needling or book into the dentist, arranging these appointments, and paying for health professional services places considerable burden on the family. This is an important treatment consideration when working with BII phobic youth. The inclusion of an education session provides an opportunity to spend more time socialising the child and their family to OST approach. It needs to be emphasised to families that OST is a kick start and that continued exposure practice is essential for the month following treatment if the child is to overcome their phobia. Moreover, the session helps parents plan for future exposure practice with the therapist discussing with them medical supplies that they may need to purchase (e.g., finger pricks from the chemist) and health practitioners (e.g., doctor or physiotherapist or pathology nurse) that they made need to schedule appointments with. To maximise treatment outcomes, Ivy and her family completed 4 × 30 min e-therapy sessions to assist them with problem-solving difficulties associated with exposure and also to keep them motivated and engaged with practicing. These sessions were integral to the continued gains she made. Families should be advised that these maintenance sessions are still an active treatment component and integral for long-term treatment success.

E-therapy sessions placed a lesser burden on Ivy's family in terms of the time needed to drive to appointments in the clinic and are more cost-effective which is consistent with the OST approach. While findings relating to the benefit of parent involvement in the treatment of anxious and phobic youth are mixed (Ollendick et al. 2015), it is suggested that for BII phobia, given the potentially higher levels of parent BII-related fears and the challenges associated with planning and organising exposure practice following OST, a modified approach which includes enhanced parental involvement may be of greater benefit these youth. Future research is needed to establish whether this is indeed correct.

It is evident from the above case illustrations that considerable preparation is required before engaging in OST for BII phobia. A significant amount of time was spent ensuring that the therapist had the appropriate materials and stimuli for exposure. Coordinating families, health professionals, and being able to access medical rooms was also time-consuming. Prior to the treatment sessions, the therapist needed to find appropriate health professionals to be involved in the treatment and to train these professionals regarding their role in the session and how to appropriately interact with phobic youth. Given that during the session the child and therapist would be exposed to blood, and other bodily fluids, and be interacting with needles, it was important for them to consider the safety and ethical issues, which may arise when conducting exposure. This is also important when considering fainting or vomiting. The therapist needs to ensure that children will be safe (e.g., making sure the child will not hit their head if they faint or there is a sick bucket available, and having a cool sugary drink ready). Additionally, therapists need to prepare themselves for OST (Reuterskiold and Öst 2012) by seeking supervision if necessary, researching the child's phobic stimuli for psychoeducation (e.g., Why is blood red? Because of the protein haemoglobin) and practicing interacting with the child's phobic stimulus (e.g., have dry needling). The therapist involved in the case illustrations had multiple dry needles, finger pricks, and blood tests each session. It was important for the therapist to be able to calmly model having these medical procedures. Moreover, the therapist needed to be able to cope when the unexpected occurred. For example, if the pathology nurse had to insert the needle multiple times in order to find a vein, the therapist had to demonstrate how to handle this.

Conclusions and Implications

The efficacy of CBT treatments for BII phobic youth is currently unknown. The outcomes of large RCT for children with specific phobia have generally led to conclusions that BII represents a more difficult phobia to treat and they

have been largely excluded from participation. Further, research examining BII phobia in youth is necessary to determine whether the proposed developmental model accurately describes these children and adolescents. Several assumptions, derived from the adult literature, have been included in the model, and these assumptions need to be empirically assessed in order to provide a developmentally sensitive conceptualisation. The aforementioned case studies describe an individualised approach to OST, based upon our proposed CBT model that addresses many of the unique characteristics associated with the clinical presentation of BII phobic youth. The development of a cost-effective and efficient treatment approach that delivers rapid reductions in fear is essential for BII phobia, given that for many children (e.g., those with medical conditions), this debilitating fear may lead to accessing less than optimal healthcare and could result in long-term health consequences. It is evident that BII phobia has a highly heterogeneous presentation in youth. Because of this, the underlying mechanisms maintaining each child's phobia may vary considerably and hence children may respond differently to different treatments. An individualised and formulation informed approach to delivering OST is therefore recommended.

Compliance with Ethical Standards

Conflict of interest The authors have no potential conflicts of interest to disclose.

References

- American Psychiatric Association (APA). (2013). *Diagnostic and statistical manual of mental disorders, 5th edition (DSM-5)*. Washington, DC: American Psychiatric Association Task Force.
- Antony, M. M., Brown, T. A., & Barlow, D. H. (1997). Heterogeneity among specific phobia types in DSM-IV. *Behaviour Research and Therapy*, 35(12), 1089–1100.
- Antony, M. M., & Watling, M. A. (2006). *Overcoming medical phobias: How to conquer fear of blood, needles, doctors, and dentists*. Oakland, CA: New Harbinger Publications.
- Armstrong, T., Hemminger, A., & Olatunji, B. O. (2013). Attentional bias in injection phobia: Overt components, time course, and relation to behavior. *Behaviour Research and Therapy*, 51(6), 266–273. doi:10.1016/j.brat.2013.02.008.
- Arntz, A., van Eck, M., & Heijmans, M. (1990). Predictions of dental pain: the fear of any expected evil, is worse than the evil itself. *Behaviour Research and Therapy*, 28(1), 29–41.
- Ayala, E. S., Meuret, A. E., & Ritz, T. (2009). Treatments for blood-injury-injection phobia: A critical review of current evidence. *Journal of Psychiatric Research*, 43(15), 1235–1242. doi:10.1016/j.jpsychires.2009.04.008.
- Bar-Haim, Y., Lamy, D., Pergamin, L., Bakermans-Kranenburg, M. J., & Van Ijzendoorn, M. H. (2007). Threat-related attentional bias in anxious and nonanxious individuals: A meta-analytic study. *Psychological Bulletin*, 133(1), 1.
- Barlow, D. H. (2004). *Anxiety and its disorders: The nature and treatment of anxiety and panic*. New York: Guilford press.

- Biederman, J., Rosenbaum, J. F., Bolduc-Murphy, E. A., Faraone, S. V., Chaloff, J., Hirshfeld, D. R., & Kagan, J. (1993). A 3-year follow-up of children with and without behavioral inhibition. *Journal of the American Academy of Child and Adolescent Psychiatry*, 32(4), 814–821. doi:10.1097/00004583-199307000-00016.
- Biederman, J., Rosenbaum, J. F., Hirshfeld, D. R., Faraone, S. V., Bolduc, E. A., Gersten, M., et al. (1990). Psychiatric correlates of behavioral inhibition in young children of parents with and without psychiatric disorders. *Archives of General Psychiatry*, 47(1), 21–26.
- Bienvenu, O. J., & Eaton, W. W. (1998). The epidemiology of blood-injection-injury phobia. *Psychological Medicine*, 28(05), 1129–1136.
- Bögels, S. M., & Brechman-Toussaint, M. L. (2006). Family issues in child anxiety: Attachment, family functioning, parental rearing and beliefs. *Clinical Psychology Review*, 26(7), 834–856. doi:10.1016/j.cpr.2005.08.001.
- Breinholdt, S., Esbjørn, B. H., Reinholdt-Dunne, M. L., & Stallard, P. (2012). CBT for the treatment of child anxiety disorders: A review of why parental involvement has not enhanced outcomes. *Journal of Anxiety Disorders*, 26(3), 416–424. doi:10.1016/j.janxdis.2011.12.014.
- Burstein, M., Georgiades, K., He, J.-P., Schmitz, A., Feig, E., Khazanov, G. K., & Merikangas, K. (2012). Specific phobia among US adolescents: Phenomenology and typology. *Depression and Anxiety*, doi:10.1002/da.22008.
- Byrne, S. P., Rapee, R. M., Malhi, G., Sweller, N., & Hudson, J. (2014). *An examination of harm beliefs in children with dog phobia*. Thesis Submitted as part of a Doctor of Philosophy, Macquarie University.
- Caseras, X., Giampietro, V., Lamas, A., Brammer, M., Vilarroya, O., Carmona, S., et al. (2010). The functional neuroanatomy of blood-injection-injury phobia: A comparison with spider phobics and healthy controls. *Psychological Medicine*, 40(1), 125–134. doi:10.1017/s0033291709005972.
- Chambless, D. L., Baker, M. J., Baucom, D. H., Beutler, L. E., Calhoun, K. S., Crits-Christoph, P., et al. (1998). Update on empirically validated therapies, II. *Clinical Psychologist*, 51(1), 3–16.
- Chambless, D. L., & Ollendick, T. H. (2001). Empirically supported psychological interventions: Controversies and evidence. *Annual Review of Psychology*, 52(1), 685–716. doi:10.1146/annurev.psych.52.1.685.
- Cisler, J. M., & Koster, E. H. W. (2010). Mechanisms of attentional biases towards threat in the anxiety disorders: An integrative review. *Clinical Psychology Review*, 30(2), 203. doi:10.1016/j.cpr.2009.11.003.
- Connolly, J., Hallam, R. S., & Marks, I. M. (1976). Selective association of fainting with blood-injury-illness fear. *Behavior Therapy*, 7(1), 8–13. doi:10.1016/s0005-7894(76)80214-6.
- Cowart, M. J. W., & Ollendick, T. H. (2013). Specific Phobias. In C. A. Essau & T. H. Ollendick (Eds.), *The treatment of childhood and adolescent anxiety* (pp. 353–368). West Sussex: Wiley.
- Dadds, M. R., Barrett, P. M., Rapee, R. M., & Ryan, S. (1996). Family process and child anxiety and aggression: An observational analysis. *Journal of Abnormal Child Psychology*, 24(6), 715–734. doi:10.1007/bf01664736.
- Davis, T. E. I. I., Ollendick, T. H., Reuther, E. T., & Muson, M. S. (2012). One-session treatment: Principles and procedures with children and adolescents. In T. E. Davis III, T. H. Ollendick, & L. G. Öst (Eds.), *Intensive one-session treatment of specific phobias* (pp. 19–42). New York: Springer.
- de Jong, P. J., & Merckelbach, H. (1998). Blood-injection-injury phobia and fear of spiders: Domain specific individual differences in disgust sensitivity. *Personality and Individual Differences*, 24(2), 153–158. doi:10.1016/S0191-8869(97)00178-5.
- de Jong, P. J., & Muris, P. (2002). Spider phobia: Interaction of disgust and perceived likelihood of involuntary physical contact. *Journal of Anxiety Disorders*, 16(1), 51–65. doi:10.1016/S0887-6185(01)00089-5.
- de Jongh, A., Muris, P., Schoenmakers, N., & Horst, G. T. (1995). Negative cognitions of dental phobics: Reliability and validity of the Dental Cognitions Questionnaire. *Behaviour Research and Therapy*, 33(5), 507–515. doi:10.1016/0005-7967(94)00081-T.
- Del Casale, A., Ferracuti, S., Rapinesi, C., Serata, D., Piccirilli, M., Savoja, V., et al. (2012). Functional neuroimaging in specific phobia. *Psychiatry Research: Neuroimaging*, 202(3), 181–197.
- Essau, C. A., Conradt, J., & Petermann, F. (2000). Frequency, comorbidity, and psychosocial impairment of specific phobia in adolescents. *Journal of Clinical Child Psychology*, 29(2), 221–231. doi:10.1207/S15374424jccp2902_8.
- Farrell, L. J., Waters, A. M., Milliner, E. L., Zimmer-Gembeck, M., McConnell, H., Tiralongo, E. et al. (2013). *D-Cycloserine augmented one session treatment of pediatric specific phobia*. Paper presented at the 7th World Congress of Behavioural and Cognitive Therapies, Lima, Peru.
- Flatt, N., & King, N. J. (2010). Brief psycho-social interventions in the treatment of specific childhood phobias: A controlled trial and a 1-year follow-up. *Behaviour Change*, 27(3), 130–153. doi:10.1375/bech.27.3.130.
- Fox, E., Russo, R., Bowles, R., & Dutton, K. (2001). Do threatening stimuli draw or hold visual attention in subclinical anxiety? *Journal of Experimental Psychology: General*, 130(4), 681–700. doi:10.1037/0096-3445.130.4.681.
- Fredrikson, M., Annas, P., & Wik, G. (1997). Parental history, aversive exposure and the development of snake and spider phobia in women. *Behaviour Research and Therapy*, 35(1), 23–28. doi:10.1016/s0005-7967(96)00076-9.
- Fyer, A. J., Mannuzza, S., Chapman, T. F., Martin, L. Y., & Klein, D. F. (1995). Specificity in familial aggregation of phobic disorders. *Archives of General Psychiatry*, 52(7), 564–573.
- Fyer, A. J., Mannuzza, S., Gallops, M. S., Martin, L. Y., Aaronson, C., Gorman, J., et al. (1990). Familial transmission of simple phobias and fears: A preliminary report. *Archives of General Psychiatry*, 47(3), 252–256. doi:10.1001/archpsyc.1990.01810150052009.
- Gar, N. S., & Hudson, J. L. (2009). The association between maternal anxiety and treatment outcome for childhood anxiety disorders. *Behaviour Change*, 26(01), 1–15. doi:10.1375/bech.26.1.1.
- Gilchrist, P. T., McGovern, G. E., Bekkouche, N., Bacon, S. L., & Ditto, B. (2015). The vasovagal response during confrontation with blood-injury-injection stimuli: The role of perceived control. *Journal of Anxiety Disorders*, 31, 43–48.
- Ginsburg, G. S., Siqueland, L., Masia-Warner, C., & Hedtko, K. A. (2004). Anxiety disorders in children: Family matters. *Cognitive and Behavioral Practice*, 11(1), 28–43. doi:10.1016/S1077-7229(04)80005-1.
- Gullone, E. (2000). The development of normal fear: A century of research. *Clinical Psychology Review*, 20(4), 429–451. doi:10.1016/S0272-7358(99)00034-3.
- Haberkamp, A., & Schmidt, T. (2014). Enhanced visuomotor processing of phobic images in blood-injury-injection fear. *Journal of Anxiety Disorders*, 28(3), 291–300. doi:10.1016/j.janxdis.2014.02.001.
- Hellström, K., Fellenius, J., & Öst, L. G. (1996). One versus five sessions of applied tension in the treatment of blood phobia. *Behaviour Research and Therapy*, 34(2), 101–112. doi:10.1016/0005-7967(95)00060-7.
- Hudson, J. L., & Rapee, R. M. (2001). Parent-child interactions and anxiety disorders: An observational study. *Behaviour Research and Therapy*, 39(12), 1411–1427. doi:10.1016/S0005-7967(00)0107-8.

- Hudson, J. L., & Rapee, R. M. (2002). Parent–child interactions in clinically anxious children and their siblings. *Journal of Clinical Child and Adolescent Psychology*, 31(4), 548–555. doi:10.1207/S15374424JCCP3104_13.
- Jones, M. K., & Menzies, R. G. (2000). Danger expectancies, self-efficacy and insight in spider phobia. *Behaviour Research and Therapy*, 38(6), 585–600. doi:10.1016/S0005-7967(99)00076-5.
- Kagan, J., Reznick, J. S., Clarke, C., Snidman, N., & Garcia-Coll, C. (1984). Behavioral inhibition to the unfamiliar. *Child Development*, 55, 2212–2225.
- Kane, E. J., Braunstein, K., Ollendick, T. H., & Muris, P. (2014). Relations of anxiety sensitivity, control beliefs, and maternal over-control to fears in clinic-referred children with specific phobia. *Journal of Child and Family Studies*. doi:10.1007/s10826-014-0014-5.
- King, N. J., Clowes-Hollins, V., & Ollendick, T. H. (1997). The etiology of childhood dog phobia. *Behaviour Research and Therapy*, 35(1), 77. doi:10.1016/S0005-7967(96)00067-8.
- King, N. J., Muris, P., & Ollendick, T. H. (2004). Specific Phobias. In T. L. Morris & J. S. March (Eds.), *Anxiety disorders in children and adolescents* (2nd ed., pp. 263–279). New York: Guilford Press.
- LeBaron, S., & Zelter, L. (1984). Assessment of acute pain and anxiety in children and adolescents by self-reports, observer reports, and a behavior checklist. *Journal of Consulting and Clinical Psychology*, 52(5), 729.
- Leutgeb, V., Schäfer, A., Köchel, A., & Schienle, A. (2012). Exposure therapy leads to enhanced late frontal positivity in 8- to 13-year-old spider phobic girls. *Biological Psychology*, 90(1), 97–104. doi:10.1016/j.biopsycho.2012.02.008.
- Leutgeb, V., & Schienle, A. (2012). Changes in facial electromyographic activity in spider-phobic girls after psychotherapy. *Journal of Psychiatric Research*, 46(6), 805–810. doi:10.1016/j.jpsychires.2012.02.017.
- Lipsitz, J. D., Barlow, D. H., Mannuzza, S., Hofmann, S. G., & Fyer, A. J. (2002). Clinical features of four DSM-IV-specific phobia subtypes. *The Journal of Nervous and Mental Disease*, 190(7), 471–478.
- Lumley, M. A., & Melamed, B. G. (1992). Blood phobics and nonphobics: Psychological differences and affect during exposure. *Behaviour Research and Therapy*, 30(5), 425–434. doi:10.1016/0005-7967(92)90026-D.
- Marks, I. (1988). Blood-injury phobia: A review. *American Journal of Psychiatry*, 145(10), 1207–1213.
- Marshall, W. L., Bristol, D., & Barbaree, H. E. (1992). Cognitions and courage in the avoidance behavior of acrophobics. *Behaviour Research and Therapy*, 30(5), 463–470. doi:10.1016/0005-7967(92)90030-K.
- Menzies, R. G., & Clarke, J. C. (1995). The etiology of phobias: A nonassociative account. *Clinical Psychology Review*, 15(1), 23–48. doi:10.1016/0272-7358(94)00039-5.
- Menzies, R. G., Harris, L. M., & Jones, M. K. (1998). Evidence from three fearful samples for a poor insight type in specific phobia. *Depression and Anxiety*, 8(1), 29–32. doi:10.1002/(SICI)1520-6394(1998)8:1<29::AID-DA6>3.0.CO;2-O.
- Merckelbach, H., Muris, P., & Schouten, E. (1996). Pathways to fear in spider phobic children. *Behaviour Research and Therapy*, 34(11–12), 935–938.
- Milliner, E. L., & Farrell, L. J. (2014). Intensive cognitive behavioural treatment for specific phobia in children and adolescents. *Psychopathology Review*, 1, 175–181.
- Milliner, E. L., Farrell, L. J., & Ollendick, T. H. (2013). Phobic anxiety. In P. Graham & S. Reynolds (Eds.), *Cognitive behaviour therapy for children and families* (3rd ed., pp. 255–274). Cambridge: Cambridge University Press.
- Mogg, K., & Bradley, B. P. (2006). Time course of attentional bias for fear-relevant pictures in spider-fearful individuals. *Behaviour Research and Therapy*, 44(9), 1241–1250. doi:10.1016/j.brat.2006.05.003.
- Mogg, K., Bradley, B. P., Miles, F., & Dixon, R. (2004). BRIEF REPORT time course of attentional bias for threat scenes: Testing the vigilance–avoidance hypothesis. *Cognition and Emotion*, 18(5), 689–700. doi:10.1080/02699930341000158.
- Moretz, M. W., Rogove, J., & McKay, D. (2011). The role of disgust in childhood anxiety disorders. In D. McKay & E. A. Storch (Eds.), *Handbook of child and adolescent anxiety disorders* (pp. 155–170). New York: Springer.
- Muris, P., & Field, A. P. (2008). Distorted cognition and pathological anxiety in children and adolescents. *Cognition and Emotion*, 22(3), 395–421. doi:10.1080/02699930701843450.
- Muris, P., & Merckelbach, H. (2001). The etiology of childhood specific phobia: A multifactorial model. In M. W. Vasey & M. R. Dadds (Eds.), *The developmental psychopathology of anxiety* (pp. 355–385). New York: Oxford University Press.
- Muris, P., Merckelbach, H., de Jong, P. J., & Ollendick, T. H. (2002). The etiology of specific fears and phobias in children: A critique of the non-associative account. *Behaviour Research and Therapy*, 40(2), 185–195. doi:10.1016/S0005-7967(01)00051-1.
- Muris, P., Merckelbach, H., Gadet, B., & Moulaert, V. (2000). Fears, worries, and scary dreams in 4- to 12-year-old children: Their content, developmental pattern, and origins. *Journal of Clinical Child Psychology*, 29(1), 43–52. doi:10.1207/S15374424jccp2901_5.
- Muris, P., Merckelbach, H., Holdrinet, I., & Sijsenaar, M. (1998). Treating phobic children: Effects of EMDR versus exposure. *Journal of Consulting and Clinical Psychology*, 66(1), 193–198. doi:10.1037/0022-006X.66.1.193.
- Muris, P., Merckelbach, H., Schmidt, H., & Tierney, S. (1999a). Disgust sensitivity, trait anxiety and anxiety disorders symptoms in normal children. *Behaviour Research and Therapy*, 37(10), 953–961. doi:10.1016/S0005-7967(99)00045-5.
- Muris, P., Merckelbach, H., Van Haften, H., & Mayer, B. (1997). Eye movement desensitisation and reprocessing versus exposure in vivo: A single-session crossover study of spider-phobic children. *The British Journal of Psychiatry*, 171(1), 82–86. doi:10.1192/bjp.171.1.82.
- Muris, P., Merckelbach, H., Wessel, I., & van de Ven, M. (1999b). Psychopathological correlates of self-reported behavioural inhibition in normal children. *Behaviour Research and Therapy*, 37(6), 575–584. doi:10.1016/S0005-7967(98)00155-7.
- Muris, P., van der Heiden, S., & Rassin, E. (2008). Disgust sensitivity and psychopathological symptoms in non-clinical children. *Journal of Behavior Therapy and Experimental Psychiatry*, 39(2), 133–146. doi:10.1016/j.jbtep.2007.02.001.
- Oar, E. L., Farrell, L. J., Waters, A. M., & Ollendick, T. H. (submitted). Blood-injection-injury phobia and dog phobia in youth: Psychological characteristics and associated features in a clinical sample. *Behavior Therapy*.
- Olatunji, B. O., & Cisler, J. M. (2009). Disgust sensitivity: Psychometric overview and operational definition. In B. O. Olatunji & D. McKay (Eds.), *Disgust and its disorders: Theory, assessment, and treatment implications*. Washington, DC: American Psychological Association.
- Olatunji, B. O., Lohr, J. M., Sawchuk, C. N., & Patten, K. (2007a). Fear and disgust responding to heterogeneous blood–injection–injury stimuli: Distinctions from anxiety symptoms. *Journal of Psychopathology and Behavioral Assessment*, 29(1), 1–8.
- Olatunji, B. O., Smits, J. A. J., Connolly, K., Willems, J., & Lohr, J. M. (2007b). Examination of the decline in fear and disgust during exposure to threat-relevant stimuli in blood–injection–

- injury phobia. *Journal of Anxiety Disorders*, 21(3), 445–455. doi:[10.1016/j.janxdis.2006.05.001](https://doi.org/10.1016/j.janxdis.2006.05.001).
- Ollendick, T. H., Halldorsdottir, T., Fraire, M. G., Austin, K. E., Noguchi, R. J. P., Lewis, K. M., et al. (2015). Specific phobias in youth: A randomized controlled trial comparing one-session treatment to a parent-augmented one-session treatment. *Behavior Therapy*, 46, 141–155.
- Ollendick, T. H., & Muris, P. (2015). The scientific legacy of Little Hans and Little Albert: Future direction for research on specific phobias in youth. *Journal of Clinical Child and Adolescent Psychology*, 44(4), 689–706.
- Ollendick, T. H., Öst, L. G., Reuterskiöld, L., Costa, N., Cederlund, R., Sirbu, C., et al. (2009). One-session treatment of specific phobias in youth: A randomized clinical trial in the United States and Sweden. *Journal of Consulting and Clinical Psychology*, 77(3), 504–516. doi:[10.1037/a0015158](https://doi.org/10.1037/a0015158).
- Ollendick, T. H., Raishevich, N., Davis, T. E. I. I., Sirbu, C., & Öst, L. G. (2010). Specific phobia in youth: Phenomenology and psychological characteristics. *Behavior Therapy*, 41(1), 133–141. doi:[10.1016/j.beth.2009.02.002](https://doi.org/10.1016/j.beth.2009.02.002).
- Öst, L. G. (1989). One-session treatment for specific phobias. *Behaviour Research and Therapy*, 27(1), 1–7. doi:[10.1016/0005-7967\(89\)90113-7](https://doi.org/10.1016/0005-7967(89)90113-7).
- Öst, L. G. (1991). Acquisition of blood and injection phobia and anxiety response patterns in clinical patients. *Behaviour Research and Therapy*, 29(4), 323–332. doi:[10.1016/0005-7967\(91\)90067-d](https://doi.org/10.1016/0005-7967(91)90067-d).
- Öst, L. G. (1992). Blood and injection phobia: Background and cognitive, physiological, and behavioral variables. *Journal of Abnormal Psychology*, 101(1), 68–74.
- Öst, L. G., Brandberg, M., & Alm, T. (1997). One versus five sessions of exposure in the treatment of flying phobia. *Behaviour Research and Therapy*, 35(11), 987–996. doi:[10.1016/S0005-7967\(97\)00077-6](https://doi.org/10.1016/S0005-7967(97)00077-6).
- Öst, L. G., Fellenius, J., & Sterner, U. (1991a). Applied tension, exposure in vivo, and tension-only in the treatment of blood phobia. *Behaviour Research and Therapy*, 29(6), 561–574. doi:[10.1016/0005-7967\(91\)90006-o](https://doi.org/10.1016/0005-7967(91)90006-o).
- Öst, L. G., & Hellström, K. (1997). Blood–Injection–Injection Phobia. In G. C. L. Davey (Ed.), *Phobias: A handbook of theory, research, and treatment* (pp. 63–80). Chichester: Wiley.
- Öst, L. G., Hellström, K., & Kåver, A. (1992). One versus five sessions of exposure in the treatment of injection phobia. *Behavior Therapy*, 23(2), 263–281. doi:[10.1016/S0005-7894\(05\)80385-5](https://doi.org/10.1016/S0005-7894(05)80385-5).
- Öst, L. G., Lindahl, I. L., Sterner, U., & Jerremalm, A. (1984a). Exposure in vivo versus applied relaxation in the treatment of blood phobia. *Behaviour Research and Therapy*, 22(3), 205–216. doi:[10.1016/0005-7967\(84\)90001-9](https://doi.org/10.1016/0005-7967(84)90001-9).
- Öst, L. G., Salkovskis, P. M., & Hellström, K. (1991b). One-session therapist-directed exposure versus self-exposure in the treatment of spider phobia. *Behavior Therapy*, 22(3), 407–422. doi:[10.1016/S0005-7894\(05\)80374-0](https://doi.org/10.1016/S0005-7894(05)80374-0).
- Öst, L. G., & Sterner, U. (1987). A specific behavioral method for treatment of blood phobia. *Behaviour Research and Therapy*, 25(1), 25–29. doi:[10.1016/0005-7967\(87\)90111-2](https://doi.org/10.1016/0005-7967(87)90111-2).
- Öst, L. G., Sterner, U., & Fellenius, J. (1989). Applied tension, applied relaxation, and the combination in the treatment of blood phobia. *Behaviour Research and Therapy*, 27(2), 109–121. doi:[10.1016/0005-7967\(89\)90069-7](https://doi.org/10.1016/0005-7967(89)90069-7).
- Öst, L. G., Sterner, U., & Lindahl, I. L. (1984b). Physiological responses in blood phobics. *Behaviour Research and Therapy*, 22(2), 109–117. doi:[10.1016/0005-7967\(84\)90099-8](https://doi.org/10.1016/0005-7967(84)90099-8).
- Öst, L. G., Svensson, L., Hellstrom, K., & Lindwall, R. (2001). One-session treatment of specific phobias in youths: a randomized clinical trial. *Journal of Consulting and Clinical Psychology*, 69(5), 814–824.
- Page, A. C. (1994). Blood–injury phobia. *Clinical Psychology Review*, 14(5), 443–461. doi:[10.1016/0272-7358\(94\)90036-1](https://doi.org/10.1016/0272-7358(94)90036-1).
- Page, A. C., & Martin, N. G. (1998). Testing a genetic structure of blood–injury–injection fears. *American Journal of Medical Genetics*, 81(5), 377–384.
- Page, A. C., & Tan, B. J. (2009). Disgust and blood–injury–injection phobia. In B. O. Olatunji & D. McKay (Eds.), *Disgust and its disorders: Theory, assessment, and treatment implications* (pp. 191–209). Washington, DC: American Psychological Association.
- Phillips, M. L., Fahy, T., David, A. S., & Senior, C. (1998). Disgust: The forgotten emotion of psychiatry. *The British Journal of Psychiatry*, 172, 373–375. doi:[10.1192/bjp.172.5.373](https://doi.org/10.1192/bjp.172.5.373).
- Rachman, S. (1976). The passing of the two-stage theory of fear and avoidance: Fresh possibilities. *Behaviour Research and Therapy*, 14(2), 125–131. doi:[10.1016/0005-7967\(76\)90066-8](https://doi.org/10.1016/0005-7967(76)90066-8).
- Rachman, S. (1977). The conditioning theory of fear acquisition: A critical examination. *Behaviour Research and Therapy*, 15(5), 375–387. doi:[10.1016/0005-7967\(77\)90041-9](https://doi.org/10.1016/0005-7967(77)90041-9).
- Rachman, S., & Cuk, M. (1992). Fearful distortions. *Behaviour Research and Therapy*, 30(6), 583–589. doi:[10.1016/0005-7967\(92\)90003-Y](https://doi.org/10.1016/0005-7967(92)90003-Y).
- Reuterskiöld, L., & Öst, L. G. (2012). Real world applications of one-session treatment. In T. E. Davis III, T. H. Ollendick, & L. G. Öst (Eds.), *Intensive one-session treatment of specific phobias* (pp. 127–141). New York: Springer.
- Ritz, T., Meuret, A. E., & Ayala, E. S. (2010). The psychophysiology of blood–injection–injury phobia: Looking beyond the diphasic response paradigm. *International Journal of Psychophysiology*, 78(1), 50–67. doi:[10.1016/j.ijpsycho.2010.05.007](https://doi.org/10.1016/j.ijpsycho.2010.05.007).
- Rudy, B. M., & Davis, T. E. I. I. (2012). Interventions for specific phobia in special populations. In T. E. Davis III, T. H. Ollendick, & L. G. Öst (Eds.), *Intensive one-session treatment of specific phobias* (pp. 19–42). New York: Springer.
- Sanders, M. R., & Jones, L. (1990). Behavioural treatment of injection, dental and medical phobias in adolescents: A case study. *Behavioural and Cognitive Psychotherapy*, 18(04), 311–316. doi:[10.1017/S0141347300010417](https://doi.org/10.1017/S0141347300010417).
- Sawchuk, C. N., Lohr, J. M., Lee, T. C., & Tolin, D. F. (1999). Exposure to disgust-evoking imagery and information processing biases in blood–injection–injury phobia. *Behaviour Research and Therapy*, 37(3), 249–257.
- Sawchuk, C. N., Lohr, J. M., Tolin, D. F., Lee, T. C., & Kleinknecht, R. A. (2000). Disgust sensitivity and contamination fears in spider and blood–injection–injury phobias. *Behaviour Research and Therapy*, 38(8), 753–762.
- Sawchuk, C. N., Lohr, J. M., Westendorf, D. H., Meunier, S. A., & Tolin, D. F. (2002). Emotional responding to fearful and disgusting stimuli in specific phobics. *Behaviour Research and Therapy*, 40(9), 1031–1046.
- Severeijns, R., Vlaeyen, J. W., van den Hout, M. A., & Weber, W. E. (2001). Pain catastrophizing predicts pain intensity, disability, and psychological distress independent of the level of physical impairment. *Clinical Journal of Pain*, 17(2), 165–172.
- Silverman, W. K., & Albano, A. M. (1996). *The anxiety disorders interview schedule for DSM-IV-child and parent versions*. London: Oxford University Press.
- Silverman, W. K., Kurtines, W. M., Ginsburg, G. S., Weems, C. F., Rabian, B., & Serafini, L. T. (1999). Contingency management, self-control, and education support in the treatment of childhood phobic disorders: A randomized clinical trial. *Journal of Consulting and Clinical Psychology*, 67(5), 675–687.
- Smith, N. B., & Meuret, A. E. (2012). The role of painful events and pain perception in blood–injection–injury fears. *Journal of*

- Behavior Therapy and Experimental Psychiatry*, 43(4), 1045–1048. doi:[10.1016/j.jbtep.2012.03.006](https://doi.org/10.1016/j.jbtep.2012.03.006).
- Taddio, A., Ipp, M., Thivakaran, S., Jamal, A., Parikh, C., Smart, S., et al. (2012). Survey of the prevalence of immunization non-compliance due to needle fears in children and adults. *Vaccine*, 30(32), 4807–4812. doi:[10.1016/j.vaccine.2012.05.011](https://doi.org/10.1016/j.vaccine.2012.05.011).
- Thompson, A. (1999). Cognitive-behavioural treatment of blood–injury–injection phobia: A case study. *Behaviour Change*, 16(03), 182–190. doi:[10.1375/bech.16.3.182](https://doi.org/10.1375/bech.16.3.182).
- Thyer, B. A., Himle, J., & Curtis, G. C. (1985). Blood–injury–illness phobia: A review. *Journal of Clinical Psychology*, 41(4), 451–459. doi:[10.1002/1097-4679\(198507\)41:4<451::aid-jclp2270410402>3.0.co;2-o](https://doi.org/10.1002/1097-4679(198507)41:4<451::aid-jclp2270410402>3.0.co;2-o).
- Tolin, D. F., Lohr, J. M., Sawchuk, C. N., & Lee, T. C. (1997). Disgust and disgust sensitivity in blood–injection–injury and spider phobia. *Behaviour Research and Therapy*, 35(10), 949–953. doi:[10.1016/S0005-7967\(97\)00048-X](https://doi.org/10.1016/S0005-7967(97)00048-X).
- Van Houtem, C. M. H. H., Laine, M. L., Boomsma, D. I., Ligthart, L., van Wijk, A. J., & De Jongh, A. (2013). A review and meta-analysis of the heritability of specific phobia subtypes and corresponding fears. *Journal of Anxiety Disorders*, 27(4), 379–388. doi:[10.1016/j.janxdis.2013.04.007](https://doi.org/10.1016/j.janxdis.2013.04.007).
- Waters, A. M., Bradley, B. P., & Mogg, K. (2014a). Biased attention to threat in paediatric anxiety disorders (generalized anxiety disorder, social phobia, specific phobia, separation anxiety disorder) as a function of ‘distress’ versus ‘fear’ diagnostic categorization. *Psychological Medicine*, 44(03), 607–616. doi:[10.1017/S0033291713000779](https://doi.org/10.1017/S0033291713000779).
- Waters, A. M., Farrell, L. J., Zimmer-Gembeck, M., Milliner, E. L., Tiralongo, E., Donovan, C. L. et al. (2014b). Augmenting one session treatment of children’s specific phobias with attention training towards positive stimuli. *Behaviour Research and Therapy*, 62, 107–119.
- Weems, C. F., & Silverman, W. K. (2006). An integrative model of control: Implications for understanding emotion regulation and dysregulation in childhood anxiety. *Journal of Affective Disorders*, 91(2), 113–124. doi:[10.1016/j.jad.2006.01.009](https://doi.org/10.1016/j.jad.2006.01.009).
- Weems, C. F., Silverman, W. K., Rapee, R. M., & Pina, A. A. (2003). The role of control in childhood anxiety disorders. *Cognitive Therapy and Research*, 27(5), 557–568. doi:[10.1023/A:1026307121386](https://doi.org/10.1023/A:1026307121386).
- Wei, C., & Kendall, P. C. (2014). Parental involvement: Contribution to childhood anxiety and its treatment. *Clinical Child and Family Psychology Review*, 17(4), 319–339. doi:[10.1007/s10567-014-0170-6](https://doi.org/10.1007/s10567-014-0170-6).

Copyright of Clinical Child & Family Psychology Review is the property of Springer Science & Business Media B.V. and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.